Covariance-Aware Policy Optimization

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Covariance-Aware Policy Optimization (CAPO) is a hierarchical Bayesian gradient aggregation method for Reinforcement Learning with Verifiable Rewards (RLVR) that replaces token-independence assumptions with a principled, data-driven treatment of structured covariance. Starting from a generalized Bayesian working model over per-trajectory gradient statistics, we place a Jeffreys prior on the scale parameter and show that the resulting posterior-optimal correction recovers the ΔL normalization as a special case. CAPO then generalizes this view by estimating an empirical variance function that captures realistic within-trajectory dependencies (e.g., banded structure, geometrically decaying, or state-dependent correlations). Under these models, trajectory-level gradient variance grows approximately linearly with length, clarifying why naive length-proportional weighting can be unstable and motivating sublinear dampening $w(L) \propto L^{-\alpha}$ with $\alpha < 1$ to control posterior risk and hedge variance-estimation noise. We provide consistent variance estimators from rollouts, a plug-in weighting rule with closed-form updates, robustness guarantees for plug-in inverse-variance weighting, and diagnostics for when CAPO reduces variance without introducing bias. Across synthetic and RLVR benchmarks, CAPO stabilizes training and accelerates wall-clock convergence relative to ΔL and token-independent baselines.

Keywords: Reinforcement Learning with Verifiable Rewards (RLVR); Bayesian Methods; Variance Reduction; Policy Optimization; Large Language Models; Sequence Modeling; Stochastic Optimization; Robust Training; Statistical Inference.

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Contributions

Overview. We address gradient aggregation for Reinforcement Learning with Verifiable Rewards (RLVR) under highly variable response lengths and realistic token dependencies. Our approach, *Covariance-Aware Policy Optimization (CAPO)*, unifies prior length-based heuristics under a Bayesian perspective and introduces a simple, plug-in rule that yields unbiased, lower-variance updates without relying on token-independence assumptions.

Key contributions.

- **Unified analysis.** We place existing length-normalization rules (including ΔL) on a coherent Bayesian footing via scale-invariant reasoning, clarifying the invariances and assumptions behind them.
- **Vector-valued optimality under common covariance shape.** We show that when per-trajectory covariances share a common "shape" up to a scalar factor, the inverse-variance rule remains optimal for a broad class of matrix risk functionals (Proposition 3.2), justifying scalar proxy modeling in high-dimensional policy gradients.
- **Robustness to variance misestimation.** We prove that plug-in inverse-variance weights degrade variance by at most a quadratic factor in the multiplicative error of the variance estimates (Proposition 3.7), providing a formal stability guarantee under model misspecification.
- **CAPO aggregator.** We generalize beyond independence by estimating per-trajectory variance from short-range covariance structure and combining it with sublinear length dampening ($\beta < 1$) to hedge estimation noise.
- **Theory to practice.** We show that trajectory-level gradient variance grows approximately linearly with length under broad dependency classes, motivating CAPO's form and its guidance for choosing $\beta < 1$.
- **Simple implementation.** CAPO is a drop-in change: variance estimates come from lightweight covariance summaries; weights are computed in a few lines and add negligible overhead.
- **Diagnostics and robustness.** We provide practical checks (effective sample size, weight entropy, CV ratios, residual autocorrelation) to tune hyperparameters and detect covariance misfit.
- **Empirical protocol.** We specify a minimal, controlled evaluation suite (§5) that compares CAPO against GRPO/DAPO/Dr. GRPO and ΔL under matched tokens. We report both end accuracy and stability diagnostics, and include targeted ablations (sublinear exponent and covariance window size) to isolate covariance-aware effects.

Takeaway. CAPO offers a principled, covariance-aware alternative to length-only normalization: it preserves unbiasedness, reduces variance, and delivers stable, efficient RLVR training with minimal engineering effort.

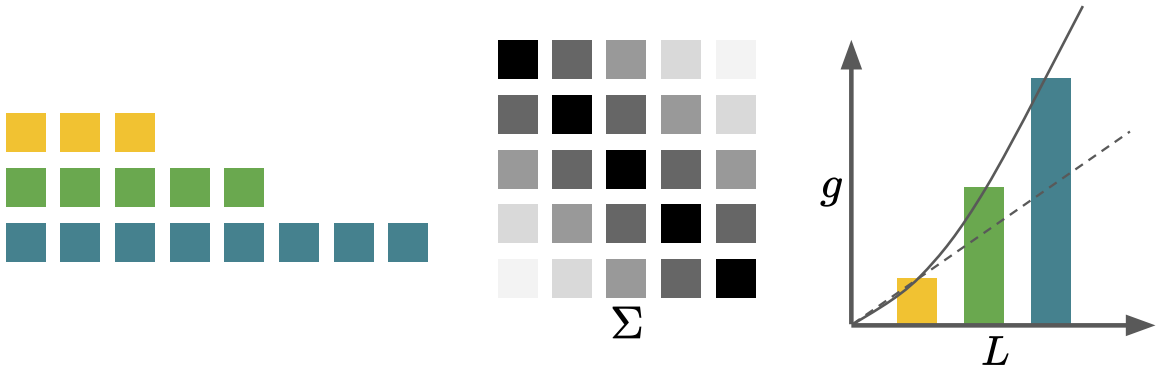

Figure 1 Schematic overview of CAPO. (a) Within-trajectory token dependence induces nontrivial covariance structure in trajectory gradients, which can amplify gradient variance as a function of response length. (b) CAPO reweights trajectories by learned posterior precisions (length and covariance-aware) to obtain an unbiased, lower-variance aggregate update.

0.1 Bayesian graphical model for CAPO

We summarize the Empirical Bayes CAPO model from §3 and §3.8 as a hierarchical Bayesian graphical model.

Generative story. At a given outer optimization step (suppressing the step index t for clarity):

1. **Hyperparameters.** Draw global parameters

$$\mu \sim p(\mu) \propto 1, \quad \sigma^2 \sim \text{Inv-Gamma}(a_0, b_0), \quad \beta \sim \pi_\beta, \quad \xi \sim \pi_\xi,$$

   where β controls the length–variance law and ξ parameterizes the dependence shape $s(L;\xi)$ (e.g., $\xi = (\rho, k, \eta)$ under the stretched-geometric k-band model of Definition 3.26).

2. **Trajectory-level structure.**

   For each trajectory $i = 1, \ldots, G$:

   (a) The trajectory length $L_i \in \mathbb{N}$ is observed (exogenous, determined by prompts and stopping rules).

   (b) Conditional on ξ and L_i, the token-level increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ form a second–order stationary process with auto-covariance $\gamma(h;\xi)$. Under the k-banded stretched-geometric family,

$$\gamma(0) = \tau^2, \qquad \gamma(h) = \tau^2 \rho^{|h|^\eta} \mathbf{1}\{|h| \le k\},$$

   so that the induced variance of the scalarized trajectory gradient $g_i = \sum_{\tau=1}^{L_i} Y_{i,\tau}$ is

$$\text{Var}(g_i \mid L_i, \xi) = \sigma^2 L_i^\beta s(L_i;\xi), \qquad s(L_i;\xi) > 0$$

   as in Lemma 3.13 and equation 16.

(c) The scalarized trajectory return $g_i \in \mathbb{R}$ obeys the heteroscedastic Normal working model

$$g_i \mid \mu, \sigma^2, \beta, \xi, L_i \sim \mathcal{N}\Big(\mu,\ v_i(\vartheta)\Big), \qquad v_i(\vartheta) = \sigma^2 L_i^{\beta} s(L_i; \xi),$$

where $\vartheta = (\beta, \xi)$.

Empirical Bayes layer and CAPO aggregator. Given observed $\{(g_i, L_i, \{Y_{i,\tau}\})\}_{i=1}^{G}$, CAPO performs Empirical Bayes (type-II ML) estimation of $\vartheta = (\beta, \xi)$ by maximizing the marginal objective $\ell(\beta, \xi)$ in Theorem 3.22. This induces precision factors

$$\omega_i(\beta, \xi) = \frac{1}{L_i^{\beta} s(L_i; \xi)}, \qquad w_i(\beta, \xi) = \frac{\omega_i(\beta, \xi)}{\sum_{j=1}^{G} \omega_j(\beta, \xi)},$$

and the CAPO aggregator

$$m(\beta, \xi) = \sum_{i=1}^{G} w_i(\beta, \xi)\, g_i,$$

which is the GLS/Bayes estimator of μ for fixed ϑ (Proposition 3.21). The weights $w_i(\beta, \xi)$ are then fed into the CAPO update as trajectory weights for the policy/teacher loss.

Plate diagram. Figure 2 shows the corresponding Bayesian plate model, highlighting the global hyperparameters $(\mu, \sigma^2, \beta, \xi)$, the trajectory-level variables (L_i, g_i), and the token-level increments $Y_{i,\tau}$ used to estimate the dependence shape via $s(L; \xi)$. The deterministic nodes $v_i(\vartheta)$, $\omega_i(\beta, \xi)$, and $m(\beta, \xi)$ summarize the variance law, precision weights, and CAPO aggregator, respectively.

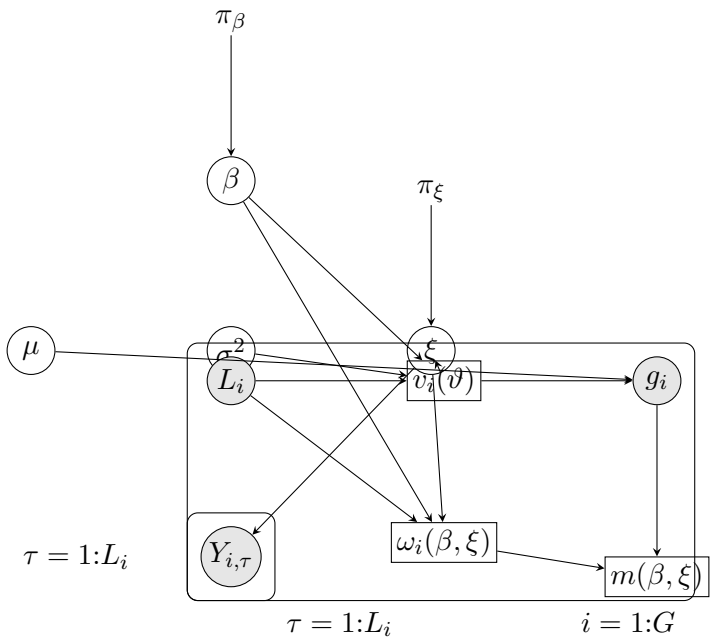

Figure 2 Bayesian plate model for CAPO with Empirical Bayes length and dependence estimation. Global hyperparameters $(\mu, \sigma^2, \beta, \xi)$ govern the heteroscedastic Normal model $g_i \mid L_i, \mu, \sigma^2, \beta, \xi \sim \mathcal{N}(\mu, \sigma^2 L_i^\beta s(L_i;\xi))$ and the token-level dependence structure of $Y_{i,\tau}$. The induced variance $v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i;\xi)$ and precision $\omega_i(\beta,\xi) = [L_i^\beta s(L_i;\xi)]^{-1}$ yield trajectory weights $w_i(\beta,\xi) \propto \omega_i(\beta,\xi)$ and the CAPO aggregator $m(\beta,\xi) = \sum_i w_i(\beta,\xi) g_i$, which is used in the policy/teacher update.

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1 Introduction

Recent work has rapidly advanced the mathematical and logical reasoning abilities of large language models (LLMs) by leveraging *reinforcement learning with verifiable rewards* (RLVR), wherein a policy produces chain-of-thought (CoT) trajectories and receives rule-based, deterministic feedback on final-answer correctness (Shao et al., 2024; Team et al., 2024; Yu et al., 2025; He et al., 2025). A distinctive challenge in RLVR is that response trajectories vary widely in length, often from tens to thousands of tokens and typically increasing as training progresses. So updates computed from a single batch can differ dramatically in both magnitude and variance (Yu et al., 2025; He et al., 2025). Consequently, training stability and sample efficiency hinge on how we aggregate per-trajectory gradients under this length variability.

A common remedy is *length-dependent* normalization. GRPO normalizes at the sample level by implicitly or explicitly dividing by trajectory length, DAPO normalizes at the batch level by the total length, and Dr. GRPO removes length terms to preserve an unbiased estimator (Shao et al., 2024; Yu et al., 2025; Liu et al., 2025). While such strategies can curb instabilities from long CoTs, their statistical properties differ: length-based factors may introduce bias in the policy-gradient direction, whereas length-agnostic updates can suffer high variance—both of which impede convergence (Williams, 1992; Sutton & Barto, 2018; Schulman et al., 2016; He et al., 2025). In particular, estimators with higher gradient variance are more prone to deviating from the optimal optimization direction, resulting in slow or unstable learning.

Related work and connections. CAPO is positioned at the intersection of three literatures.

First, it relates to modern RLHF/RLVR optimization pipelines for LLMs, which typically instantiate a policy-gradient method (most often a PPO-type update (Schulman et al., 2017)) equipped with reward modeling, preference modeling, or verifiable rewards (Ouyang et al., 2022; Shao et al., 2024; Rafailov et al., 2023). The specific normalization rules compared in this paper (ΔL, GRPO-, DAPO-, and Dr. GRPO-type variants) can be interpreted as different choices of *trajectory-level reweighting* and *baseline* in a policy-gradient estimator. Our contribution is to place these heuristics into a single estimation-theoretic framework and to isolate a design axis that is largely implicit in prior work: how *within-trajectory dependence* and heteroscedasticity induced by variable-length CoT should enter the weighting rule.

Second, CAPO has close ties to classical weighted least squares and generalized least squares. The MVU weights in Lemma 3.6 are the Gauss–Markov/BLUE solution for estimating a common mean under heteroscedastic noise, and the dependence-aware extensions can be viewed as (structured) GLS with autocorrelated errors (Hamilton, 1994). This connection motivates both the functional form of the optimal weights and the emphasis on correctly modeling how variance grows with length and dependence.

Third, our empirical-Bayes (EB) construction in §3.8 follows a long tradition of EB modeling for structured variance and shrinkage, in which hyperparameters are fitted by marginal likelihood and then plugged into a decision rule (Efron, 2012). In the present setting, EB serves a pragmatic purpose: it yields closed-form weight expressions with stable large-L behavior while avoiding the computational overhead of full Bayesian integration over dependence parameters.

This paper proposes *Covariance-Aware Policy Optimization (CAPO)*, a principled, variance-aware gradient aggregation framework for RLVR. We begin by casting per-step loss/gradient aggregation as a classical *minimum-variance unbiased estimation* (MVUE) problem over heterogeneous trajectory lengths, in the same spirit as He et al. (2025). We then introduce a generalized Bayesian working model for trajectory-level gradient statistics, place a Jeffreys prior on the scale parameter to obtain an invariant baseline, and show that the resulting posterior-precision aggregator recovers ΔL normalization as a special case (Jeffreys, 1946; He et al., 2025). Finally, we extend beyond token-independence by modeling realistic within-trajectory dependencies via simple covariance structures (e.g., banded or geometrically decaying kernels). This yields a plug-in weighting rule that down-weights trajectories by both an empirically estimated variance and a sublinear length factor:

$$\bar{g} = \sum_{i=1}^{M} \tilde{w}_i g_i, \qquad \tilde{w}_i \propto \frac{L_i^{-\alpha}}{\widehat{v}_i}, \qquad \alpha \in (0,1), \tag{1}$$

where g_i is the unnormalized gradient contribution from trajectory i, L_i is its length, and $\widehat{v}_i$ estimates the trajectory-level gradient variance under the learned covariance model.

Our analysis shows that, under broad dependence classes, the trajectory-level variance grows *approximately linearly* with length whenever within-trajectory correlations are summable (short-range dependence). For a simple but illustrative case with geometrically decaying correlations $\mathrm{Cov}(Z_t, Z_s) = \sigma^2 \rho^{|t-s|}$ (with $|\rho| < 1$), one obtains

$$\mathrm{Var}\left(\sum_{t=1}^{L} Z_t\right) = \sigma^2 \sum_{k=-(L-1)}^{L-1} (L - |k|)\, \rho^{|k|} = \Theta(L), \qquad |\rho| < 1, \tag{2}$$

so that naive length-proportional weighting can be unstable. In finite samples, additionally, variance estimation noise increases with L, which motivates the sublinear dampening in equation 1. We further study the *coefficient of variation* (CV) of the aggregated estimator under different normalization schemes and show that CAPO reduces CV relative to length-agnostic updates while avoiding the length-induced bias of sample- or batch-normalization.

Empirically, we evaluate CAPO on a focused RLVR testbed: a representative task (Count-Down), a single open-weight backbone model (Qwen2.5–1.5B–Instruct), two context-length regimes (2048 and 8192), and a compact set of aggregation baselines that isolate the effect of aggregation under matched tokens and identical decoding/clipping (GRPO Norm, DAPO Norm, Dr. GRPO Norm, ΔL, and CAPO). In addition to end performance, we report *stability diagnostics* (monotonicity of held-out accuracy, step-to-step drift, CV/kurtosis of trajectory weights, and length-decile stratification), together with a targeted ablation over the covariance bandwidth used by CAPO's empirical-Bayes estimator.

Summary of contributions. (i) A unified estimation-theoretic and Bayesian analysis of RLVR gradient aggregation that derives ΔL from MVUE/Jeffreys principles and clarifies invariances and bias–variance trade-offs (Jeffreys, 1946; He et al., 2025); (ii) *CAPO*, a covariance-aware generalization with consistent variance estimation from rollouts and closed-form plug-in weights as in equation 1; (iii) dependence-aware theory linking token-level autocovariance to trajectory-level variance growth, yielding principled corrections beyond length-only normalization; and

(iv) an empirical protocol with stability diagnostics and targeted ablations that isolate covariance-aware weighting effects under matched compute.

Organization. Section 2 formalizes the trajectory aggregation problem and clarifies the role of the scaling constraint in equation 6. Section 3 develops the MVU and Bayesian foundations, derives dependence-aware variance laws, and establishes the EB estimator and its asymptotics. Section 4 provides implementable CAPO variants and complexity considerations. Section 5 describes the RLVR evaluation protocol and stability diagnostics. Appendices supply full proofs and illustrative covariance examples.

2 Problem Setup

Goal. Given a batch of trajectories with heterogeneous lengths and within-trajectory token dependence, we seek an unbiased linear aggregate of trajectory-level gradients that is statistically efficient. We cast the aggregation problem as estimation of a common mean under heteroscedastic noise whose variance scales with trajectory length and dependence structure, setting up a Bayesian treatment that will later recover and generalize the ΔL family of weights (He et al., 2025).

2.1 Setup and Notation

We consider an update step that aggregates a collection of $G \geq 1$ rollouts (trajectories) indexed by $i \in \{1, \ldots, G\}$. Trajectory i has a (random) token length $L_i \in \mathbb{N}$ and yields a sequence of token-level gradient contributions $\{\boldsymbol{Z}_{i,t}\}_{t=1}^{L_i} \subset \mathbb{R}^p$, where p is the parameter dimension. The (un-normalized) trajectory gradient is the token-sum

$$\boldsymbol{g}_i \;:=\; \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}. \tag{3}$$

Throughout, we regard the policy-gradient signal as a common mean vector $\boldsymbol{\mu} := \mathbb{E}[\boldsymbol{g}_i]$ (under the current policy and sampling scheme), and we study linear aggregators of the form equation 6. The scaling constant $M > 0$ in equation 6 is fixed by the training objective (e.g., M may correspond to a number of prompt groups in a batch, or to a normalization convention in the policy-gradient estimator).

To quantify *dispersion* of a vector-valued gradient, we introduce a scalar functional $\Psi : \mathbb{S}_+^p \to (0, \infty)$ on positive-semidefinite matrices (e.g., $\Psi(\boldsymbol{\Sigma}) = \mathrm{tr}(\boldsymbol{\Sigma})$ or $\Psi(\boldsymbol{\Sigma}) = \boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u}$ for a fixed direction $\boldsymbol{u}$) and define the proxy variance

$$v_i \;:=\; \Psi(\mathrm{Cov}(\boldsymbol{g}_i))\,. \tag{4}$$

The role of CAPO is to estimate how v_i depends on length and within-trajectory dependence, and to use (an estimate of) v_i^{-1} as the statistically efficient weighting rule.

Assumption 2.1 (Cross-trajectory uncorrelatedness)**.** *For the trajectories aggregated in a single update, we assume* $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) = \boldsymbol{0}$ *for all* $i \neq j$*. This assumption is exact when trajectories are sampled independently and is a standard working approximation when any remaining dependence is negligible compared to within-trajectory autocorrelation.*

Symbol	Meaning
G	number of trajectories aggregated in the update
L_i	token length of trajectory i
p	parameter dimension; $\boldsymbol{Z}_{i,t} \in \mathbb{R}^p$
$\boldsymbol{Z}_{i,t}$	token-level gradient contribution at position t in trajectory i
$\boldsymbol{g}_i$	trajectory-level gradient $\sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}$
$\boldsymbol{\mu}$	common mean gradient, $\mathbb{E}[\boldsymbol{g}_i] = \boldsymbol{\mu}$
$\widehat{\boldsymbol{g}}$	aggregated gradient $\sum_{i=1}^{G} x_i \boldsymbol{g}_i$
x_i	trajectory weights; unbiasedness requires $\sum_i x_i = 1/M$
M	fixed scaling constant in equation 6
$\Psi(\cdot)$	scalar functional on covariances used to define v_i
v_i	proxy dispersion $\Psi(\mathrm{Cov}(\boldsymbol{g}_i))$
$\boldsymbol{\Gamma}_i(k)$	token-level lag-k autocovariance, equation 7
$s(L;\xi)$	dependence shape factor linking autocovariance to v_i
β	length exponent in the variance law $v_i \propto L_i^{\beta}$

Table 1 Core notation used throughout the paper.

Assumption 2.2 (Second-order stationarity within trajectories)**.** *For each trajectory i, conditional on its length L_i, the token-level process $\{\boldsymbol{Z}_{i,t}\}_{t=1}^{L_i}$ is second-order stationary in the sense that $\mathbb{E}[\boldsymbol{Z}_{i,t}]$ does not depend on t and the lag-k autocovariance $\mathrm{Cov}(\boldsymbol{Z}_{i,t}, \boldsymbol{Z}_{i,t-k})$ depends on k but not on t whenever both indices are in $\{1, \ldots, L_i\}$.*

Assumptions 2.1–2.2 are sufficient for the variance decompositions used in §3 and the appendices. We discuss several dependence models for $\boldsymbol{\Gamma}_i(k)$ and their implications for the length-variance law in §2.4 and §3.6.

2.2 Observed Entities and Target Quantity

We observe G trajectories indexed by $i \in \{1, \ldots, G\}$. Trajectory i has length $L_i \in \mathbb{N}$ tokens and produces the un-normalized trajectory-level gradient.

$$\boldsymbol{g}_i \;=\; \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t} \;\in\; \mathbb{R}^p, \qquad \mathbb{E}[\,\boldsymbol{g}_i\,] \;=\; \boldsymbol{\mu} \;\in\; \mathbb{R}^p, \tag{5}$$

where $\boldsymbol{Z}_{i,t}$ is the token-level score contribution (e.g., advantage-weighted policy-score increment). The learning signal $\boldsymbol{\mu}$ is common across trajectories in a batch.

2.3 A Statistically Efficient, Linear, Unbiased Estimator

Training aggregates the per-trajectory gradients linearly,

$$\widehat{\boldsymbol{g}} \;=\; \sum_{i=1}^{G} x_i\, \boldsymbol{g}_i, \qquad \sum_{i=1}^{G} x_i \;=\; \frac{1}{M}, \tag{6}$$

for a fixed scaling constant $M > 0$. Under equation 5, any choice of weights satisfying the normalization in equation 6 yields an unbiased estimate, $\mathbb{E}[\widehat{\boldsymbol{g}}] = \boldsymbol{\mu}/M$. The design problem is to choose $\{x_i\}$ to minimize the dispersion of $\widehat{\boldsymbol{g}}$ subject to unbiasedness.

2.4 Non-iid Tokens

Token contributions within a trajectory are dependent. Define the auto-covariance function,

$$\boldsymbol{\Gamma}_i(k) \;:=\; \mathrm{Cov}\Big(\boldsymbol{Z}_{i,t},\, \boldsymbol{Z}_{i,t-k}\Big) \in \mathbb{R}^{p\times p}, \qquad k \in \mathbb{Z}. \tag{7}$$

Then the trajectory covariance, under stationarity condition, consult Appendix C, is

$$\mathrm{Var}(\boldsymbol{g}_i) \;=\; \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k). \tag{8}$$

For aggregation we summarize $\mathrm{Var}(\boldsymbol{g}_i)$ by a positive scalar functional

$$v_i \;:=\; \Psi\Big(\mathrm{Var}(\boldsymbol{g}_i)\Big) \;\in\; (0,\infty), \qquad \text{e.g., } \Psi(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A}) \text{ or } \Psi(\boldsymbol{A}) = \mathbb{E}_{\boldsymbol{u}}[\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}], \tag{9}$$

where $\boldsymbol{u}$ is a unit vector (fixed or randomized). If $\{v_i\}$ were known, the minimum-variance unbiased (MVU) linear weights are

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^{G} v_j^{-1}}. \tag{10}$$

For a detailed proof along with the assumptions consult Appendix A. Hence the core statistical task is to model/estimate v_i as a function of length and dependence.

3 Method

Roadmap. We begin with a length-only model that recovers ΔL weighting method; we then present a heteroscedastic Gaussian model with flexible priors, derive the posterior-precision aggregator, identify Jeffreys' baseline (α=1), and show how correctness-linked variance reduction yields α<1; dependence corrections are incorporated via $s(L;\xi)$.

3.1 Length-Only Benchmark and the ΔL Family

Under an i.i.d. token model with $\mathrm{Var}(\boldsymbol{Z}_{i,t}) = \boldsymbol{\Sigma}_0$ and $\boldsymbol{\Gamma}_i(k) = \boldsymbol{0}$ for $k \neq 0$,

$$\mathrm{Var}(\boldsymbol{g}_i) = L_i\, \boldsymbol{\Sigma}_0 \quad \Longrightarrow \quad v_i \propto L_i. \tag{11}$$

Substituting into equation 10 yields the inverse-length rule

$$x_i \;\propto\; L_i^{-1}, \tag{12}$$

which is precisely the ΔL normalization of He et al. (2025) with exponent $\alpha = 1$. More generally, following He et al. (2025) we consider the one-parameter family

$$x_i \;\propto\; L_i^{-\beta}, \qquad \beta \in [0,2], \tag{13}$$

interpolating between constant weighting ($\beta = 0$) and inverse-length weighting ($\beta = 1$), as a practical control of variance induced by heterogeneous lengths.

A two-parameter normalization family. Length-only normalization can be viewed as choosing a (possibly biased) *global* step scale and a (possibly variance-reducing) *relative* within-batch reweighting. To decouple these effects, we introduce a two-parameter family

$$x_i(\alpha,\kappa) \;:=\; \frac{1}{M_\kappa}\,\frac{L_i^{-\alpha}}{\sum_{j=1}^{G} L_j^{-\alpha}}, \qquad M_\kappa \;:=\; \Big(\sum_{j=1}^{G} L_j^{-\alpha}\Big)^{\kappa}, \tag{14}$$

where $\alpha \in [0,1]$ controls the relative suppression of long responses and $\kappa \in \mathbb{R}$ controls the overall step magnitude. The standard ΔL estimator corresponds to $\kappa = 0$ (purely relative reweighting with a fixed M), while $\kappa = 1$ recovers a DAPO-style batch-length normalization where the step magnitude shrinks with the batch effective length. Section 3.8 shows that CAPO preserves unbiasedness by enforcing the linear constraint $\sum_i x_i = 1/M$ while learning the variance structure; empirically, we therefore treat (α,κ) as a diagnostic lens rather than as free hyperparameters for CAPO.

3.2 Bayesian Setup under a General Prior

We project each trajectory-level gradient onto a fixed direction to obtain a scalar summary. Let $\boldsymbol{u}$ be a unit vector and define

$$g_i \;=\; \boldsymbol{u}^\top \boldsymbol{g}_i, \qquad \mathbb{E}[g_i] = \mu, \qquad \mathrm{Var}(g_i) = v_i. \tag{15}$$

Vector-valued gradients and common-shape covariances. Our Bayesian and empirical-Bayes derivations proceed via a scalar projection equation 15. This is not merely a notational convenience: in a broad and practically relevant regime, the resulting weights are also optimal for *vector-valued* aggregation objectives.

Assumption 3.1 (Common-shape covariance)**.** *There exists a fixed positive-semidefinite matrix $\boldsymbol{\Sigma}_0 \in \mathbb{S}_+^p$ and positive scalars v_i such that, for every trajectory i,*

$$\mathrm{Cov}(\boldsymbol{g}_i) = v_i\,\boldsymbol{\Sigma}_0.$$

Proposition 3.2 (Inverse-variance weights are optimal under common shape)**.** *Assume cross-trajectory uncorrelatedness (Assumption 2.1) and the common-shape model in Assumption 3.1. Let $\Psi : \mathbb{S}_+^p \to (0,\infty)$ be any positive-homogeneous functional, i.e., $\Psi(c\boldsymbol{A}) = c\,\Psi(\boldsymbol{A})$ for all $c \geq 0$ and $\boldsymbol{A} \in \mathbb{S}_+^p$. Then the unique minimizer of $\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}}))$ over weights satisfying $\sum_{i=1}^G x_i = 1/M$ is given by the inverse-variance rule $x_i^\star \propto v_i^{-1}$ (with normalization fixed by $\sum_i x_i = 1/M$), i.e., the same weights as in Lemma 3.6.*

Proof. Under cross-trajectory uncorrelatedness,

$$\mathrm{Cov}(\widehat{\boldsymbol{g}}) = \sum_{i=1}^{G} x_i^2\,\mathrm{Cov}(\boldsymbol{g}_i) = \sum_{i=1}^{G} x_i^2\,v_i\,\boldsymbol{\Sigma}_0 = \Big(\sum_{i=1}^{G} x_i^2 v_i\Big)\,\boldsymbol{\Sigma}_0.$$

By positive homogeneity of Ψ,

$$\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}})) = \Big(\sum_{i=1}^{G} x_i^2 v_i\Big)\,\Psi(\boldsymbol{\Sigma}_0).$$

Since $\Psi(\mathbf{\Sigma}_0) > 0$ is constant in (x_i), minimizing $\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}}))$ is equivalent to minimizing $\sum_i x_i^2 v_i$ under $\sum_i x_i = 1/M$, whose unique minimizer is the inverse-variance rule in Lemma 3.6. □

Remark 3.3 (Beyond common shape). When the covariance matrices $\mathrm{Cov}(\boldsymbol{g}_i)$ are not approximately proportional to a common shape, the optimal weights can depend on the chosen scalarization objective Ψ (e.g., a particular direction $\boldsymbol{u}$ or a trace objective). CAPO focuses on the setting where a *single* scalar proxy v_i provides a stable and implementable approximation.

Conditional on the variance parameters $\vartheta := (\sigma^2, \beta, \xi)$,

$$g_i \,\Big|\, \mu, \vartheta \;\sim\; \mathcal{N}\Big(\mu,\; v_i(\vartheta)\Big), \qquad v_i(\vartheta) \;=\; \sigma^2\, L_i^{\beta}\, s(L_i;\, \xi), \tag{16}$$

where $\sigma^2 > 0$ is a global scale, $\beta \in [0, 2]$ controls length scaling, and $s(L_i; \xi) > 0$ is a shape factor encoding within-trajectory dependence. This modular treatment allows us to plug in different estimates for the shape factor neatly.

General priors. We adopt a flat prior on the mean and a flexible, proper prior on the variance and shape parameters:

$$p(\mu) \;\propto\; 1, \qquad \sigma^2 \sim \text{Inv-Gamma}(a_0, b_0), \qquad \beta \sim \pi_\beta, \qquad \xi \sim \pi_\xi. \tag{17}$$

Jeffreys' prior for scale is recovered in the limit $a_0, b_0 \to 0$. The prior π_β can be chosen agnostic (centered at 1 with moderate variance) or informative to reflect anticipated sub/superlinear growth. The prior π_ξ encodes beliefs about token dependence.

Likelihood geometry and sufficient sums. Define precisions and aggregated quantities

$$\lambda_i(\vartheta) := \frac{1}{v_i(\vartheta)} = \frac{1}{\sigma^2}\,\omega_i(\beta, \xi), \qquad \omega_i(\beta, \xi) := L_i^{-\beta}\, s(L_i; \xi)^{-1}, \tag{18}$$

$$\Lambda(\vartheta) \;=\; \sum_{i=1}^{G} \lambda_i(\vartheta), \qquad T(\vartheta) \;=\; \sum_{i=1}^{G} \lambda_i(\vartheta)\, g_i. \tag{19}$$

Then the conditional posterior for μ given ϑ is Normal:

$$\mu \,\Big|\, \vartheta, g_i \;\sim\; \mathcal{N}\Big(m(\vartheta),\; \Lambda(\vartheta)^{-1}\Big), \qquad m(\vartheta) = \frac{T(\vartheta)}{\Lambda(\vartheta)} = \frac{\sum_i \lambda_i(\vartheta) g_i}{\sum_i \lambda_i(\vartheta)}. \tag{20}$$

Let $m(\beta, \xi)$ denote $m(\vartheta)$ with σ^2 factored out via equation 18, and define the weighted residual sum of squares

$$\mathrm{RSS}_\omega(\beta, \xi) \;=\; \sum_{i=1}^{G} \omega_i(\beta, \xi)\, \Big(g_i - m(\beta, \xi)\Big)^2. \tag{21}$$

Combining equation 17–equation 21, the conditional posterior of σ^2 is

$$\sigma^2 \,\Big|\, \beta, \xi, \{g_i\} \;\sim\; \text{Inv-Gamma}\Big(a_0 + \frac{G-1}{2},\; b_0 + \frac{1}{2}\,\mathrm{RSS}_\omega(\beta, \xi)\Big). \tag{22}$$

Proposition 3.4 (Conditional posterior of σ^2)**.** *Under the Gaussian model*

$$g_i \mid \mu, \vartheta \;\sim\; \mathcal{N}\big(\mu,\; v_i(\vartheta)\big), \qquad v_i(\vartheta) = \sigma^2 L_i^{\beta} s(L_i;\xi) \;=\; \frac{\sigma^2}{\omega_i(\beta,\xi)},$$

with $\vartheta = (\sigma^2, \beta, \xi)$ *and* $\omega_i(\beta,\xi) := L_i^{-\beta} s(L_i;\xi)^{-1}$ *(cf. equation 18), and priors* $p(\mu) \propto 1$, $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$ *as in equation 17, the conditional posterior of* σ^2 *given* (β,ξ) *and data* $\{g_i\}_{i=1}^G$ *is*

$$\sigma^2 \,\Big|\, \beta, \xi, \{g_i\} \;\sim\; \text{Inv-Gamma}\Big(a_0 + \frac{G-1}{2},\; b_0 + \frac{1}{2}\,\text{RSS}_\omega(\beta,\xi)\Big), \tag{22}$$

where the weighted mean is

$$m(\beta,\xi) \;:=\; \frac{\sum_{i=1}^G \omega_i(\beta,\xi)\, g_i}{\sum_{i=1}^G \omega_i(\beta,\xi)}$$

and the weighted residual sum of squares is

$$\text{RSS}_\omega(\beta,\xi) \;:=\; \sum_{i=1}^G \omega_i(\beta,\xi)\,\Big(g_i - m(\beta,\xi)\Big)^2. \tag{21}$$

Proof. Step 1: Likelihood in (μ,σ^2) *up to* (β,ξ)*-constants.* Conditional on (β,ξ), the variances are $v_i = \sigma^2/\omega_i$. Thus the likelihood is

$$p(\{g_i\} \mid \mu, \sigma^2, \beta, \xi) = \prod_{i=1}^G \frac{\sqrt{\omega_i}}{\sqrt{2\pi\sigma^2}} \exp\Big\{-\frac{\omega_i}{2\sigma^2}(g_i-\mu)^2\Big\}.$$

Dropping factors independent of (μ,σ^2) (i.e., $\prod_i \sqrt{\omega_i}$), the kernel is

$$p(\{g_i\} \mid \mu, \sigma^2, \beta, \xi) \;\propto\; (\sigma^2)^{-G/2} \exp\Big\{-\frac{1}{2\sigma^2}\sum_{i=1}^G \omega_i (g_i-\mu)^2\Big\}. \tag{23}$$

Step 2: Multiply by priors. With $p(\mu) \propto 1$ and $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$ whose density kernel is $(\sigma^2)^{-a_0-1}\exp\{-b_0/\sigma^2\}$, the joint posterior kernel is

$$p(\mu, \sigma^2 \mid \beta, \xi, \{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)} \exp\Big\{-\frac{1}{\sigma^2}\Big(b_0 + \frac{1}{2}\sum_{i=1}^G \omega_i (g_i-\mu)^2\Big)\Big\}.$$

Step 3: Complete the square in μ*.* Let $\Lambda_\omega := \sum_{i=1}^G \omega_i$ and $m(\beta,\xi) := \big(\sum_i \omega_i g_i\big)/\Lambda_\omega$. Using the weighted-mean identity,

$$\sum_{i=1}^G \omega_i (g_i-\mu)^2 = \sum_{i=1}^G \omega_i \Big(g_i - m(\beta,\xi)\Big)^2 + \Lambda_\omega\,(\mu - m(\beta,\xi))^2,$$

we obtain

$$p(\mu, \sigma^2 \mid \beta, \xi, \{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)} \exp\Big\{-\frac{1}{\sigma^2}\Big(b_0 + \frac{1}{2}\text{RSS}_\omega(\beta,\xi)\Big)\Big\} \exp\Big\{-\frac{\Lambda_\omega}{2\sigma^2}\,(\mu - m(\beta,\xi))^2\Big\}.$$

Step 4: Integrate out μ. Because $p(\mu) \propto 1$ and the μ-kernel is Gaussian, we have

$$\int_{\mathbb{R}} \exp\left\{-\frac{\Lambda_\omega}{2\sigma^2}(\mu - m)^2\right\} d\mu = \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}},$$

which contributes a factor $(\sigma^2)^{1/2}$ (and a constant in (β, ξ)) to the kernel. Therefore,

$$p(\sigma^2 \mid \beta, \xi, \{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)}(\sigma^2)^{1/2} \exp\left\{-\frac{1}{\sigma^2}\left(b_0 + \frac{1}{2}\mathrm{RSS}_\omega(\beta, \xi)\right)\right\}.$$

Step 5: Identify the inverse-gamma form. Collecting powers of σ^2 gives

$$p(\sigma^2 \mid \beta, \xi, \{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+\frac{G-1}{2}+1\right)} \exp\left\{-\frac{1}{\sigma^2}\left(b_0 + \frac{1}{2}\mathrm{RSS}_\omega(\beta, \xi)\right)\right\},$$

which is the kernel of Inv-Gamma$\Big(a_0 + \frac{G-1}{2},\; b_0 + \frac{1}{2}\mathrm{RSS}_\omega(\beta, \xi)\Big)$. □

Remark 3.5. (i) The factorization $\lambda_i(\vartheta) = \sigma^{-2}\omega_i(\beta, \xi)$ implies $m(\beta, \xi)$ does *not* depend on σ^2, since the common factor cancels: $m(\vartheta) = T(\vartheta)/\Lambda(\vartheta) = \Big(\sum \lambda_i g_i\Big)/\Big(\sum \lambda_i\Big) = \Big(\sum \omega_i g_i\Big)/\Big(\sum \omega_i\Big)$.
(ii) The shape update $a_0 \mapsto a_0 + \frac{G-1}{2}$ reflects the loss of one degree of freedom from estimating the (flat-prior) mean μ.
(iii) Constants that depend only on (β, ξ), such as $\prod_i \sqrt{\omega_i}$ and $\Lambda_\omega^{-1/2}$, are immaterial for $p(\sigma^2 \mid \beta, \xi, \{g_i\})$ and are absorbed into the normalizing constant.

3.3 Formal Results and Aggregators

Lemma 3.6 (MVU weights with known v_i)**.** *Let $(g_1, \ldots, g_G)$ be scalar random variables such that $\mathbb{E}[g_i] = \mu$ for all i, $\mathrm{Var}(g_i) = v_i \in (0, \infty)$ are known, and the trajectories are cross-uncorrelated: $\mathrm{Cov}(g_i, g_j) = 0$ for all $i \neq j$. For any weights $(x_1, \ldots, x_G)$ satisfying the unbiasedness constraint $\sum_{i=1}^G x_i = 1/M$, the linear estimator $\widehat{\mu} := \sum_{i=1}^G x_i g_i$ is unbiased for μ/M and satisfies*

$$\mathrm{Var}(\widehat{\mu}) = \sum_{i=1}^G x_i^2 v_i.$$

The unique minimizer of $\mathrm{Var}(\widehat{\mu})$ over all such unbiased linear estimators is

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^G v_j^{-1}}, \qquad \text{(10 revisited)}$$

i.e., inverse-variance weighting with normalization fixed by $\sum_i x_i = 1/M$.

Proof. Under the stated cross-uncorrelatedness, for any weights $(x_1, \ldots, x_G)$ we have

$$\mathrm{Var}\left(\sum_{i=1}^G x_i g_i\right) = \sum_{i=1}^G x_i^2\,\mathrm{Var}(g_i) = \sum_{i=1}^G x_i^2 v_i.$$

We minimize the strictly convex quadratic function $Q(x) := \sum_{i=1}^G x_i^2 v_i$ over the affine subspace $\{x \in \mathbb{R}^G : \sum_{i=1}^G x_i = 1/M\}$.

Introduce the Lagrangian

$$\mathcal{L}(x_1, \ldots, x_G, \lambda) = \sum_{i=1}^{G} x_i^2 v_i - \lambda \left(\sum_{i=1}^{G} x_i - \frac{1}{M} \right).$$

The first-order optimality conditions are, for each $i \in \{1, \ldots, G\}$,

$$\frac{\partial \mathcal{L}}{\partial x_i} = 2x_i v_i - \lambda = 0 \quad \Longrightarrow \quad x_i = \frac{\lambda}{2v_i}.$$

Imposing the constraint $\sum_i x_i = 1/M$ yields

$$\frac{1}{M} = \sum_{i=1}^{G} x_i = \frac{\lambda}{2} \sum_{i=1}^{G} v_i^{-1} \quad \Longrightarrow \quad \frac{\lambda}{2} = \frac{1/M}{\sum_{j=1}^{G} v_j^{-1}}.$$

Substituting back gives

$$x_i^\star = \frac{1}{M} \frac{v_i^{-1}}{\sum_{j=1}^{G} v_j^{-1}},$$

which satisfies the constraint and the first-order conditions. Since Q is strictly convex (all $v_i > 0$), the solution is the unique global minimizer. □

Proposition 3.7 (Robustness of plug-in inverse-variance weights)**.** *Assume the setting of Lemma 3.6 and let $\{\widehat{v}_i\}_{i=1}^G$ be positive variance estimates such that, for some $c \geq 1$,*

$$\frac{1}{c} v_i \;\leq\; \widehat{v}_i \;\leq\; c\, v_i \qquad \text{for all } i \in \{1, \ldots, G\}. \tag{24}$$

Define the plug-in inverse-variance weights

$$\widehat{x}_i \;:=\; \frac{1}{M} \frac{\widehat{v}_i^{-1}}{\sum_{j=1}^{G} \widehat{v}_j^{-1}}.$$

Let $V^\star$ denote the minimum variance achieved by the oracle MVU weights of Lemma 3.6, and let $\widehat{V} := \mathrm{Var}(\sum_i \widehat{x}_i g_i)$ denote the variance under the plug-in weights. Then the plug-in rule is variance-robust in the sense that

$$\frac{1}{c^2} V^\star \;\leq\; \widehat{V} \;\leq\; c^2 V^\star.$$

Proof. Let $a_i := v_i^{-1}$ and $\widehat{a}_i := \widehat{v}_i^{-1}$ so that equation 24 is equivalent to $\widehat{a}_i \in [a_i/c,\, ca_i]$. Define $S := \sum_{i=1}^{G} a_i$ and $\widehat{S} := \sum_{i=1}^{G} \widehat{a}_i$. A direct computation (see the proof of Lemma 3.6) gives the oracle optimum

$$V^\star = \frac{1}{M^2} \frac{1}{S}.$$

For the plug-in weights, cross-trajectory uncorrelatedness gives

$$\widehat{V} = \sum_{i=1}^{G} \widehat{x}_i^2 v_i = \frac{1}{M^2} \frac{\sum_{i=1}^{G} \widehat{a}_i^2 v_i}{\widehat{S}^2} = \frac{1}{M^2} \frac{\sum_{i=1}^{G} \widehat{a}_i^2 / a_i}{\widehat{S}^2}.$$

Since $\widehat{a}_i/a_i \le c$, we have $\widehat{a}_i^2/a_i = \widehat{a}_i(\widehat{a}_i/a_i) \le c\,\widehat{a}_i$ and hence $\sum_i \widehat{a}_i^2/a_i \le c\,\widehat{S}$. Therefore

$$\widehat{V} \le \frac{1}{M^2}\,\frac{c}{\widehat{S}}.$$

Moreover, $\widehat{a}_i \ge a_i/c$ implies $\widehat{S} \ge S/c$, so $\widehat{V} \le \frac{1}{M^2}\frac{c}{S/c} = c^2 V^\star$.

For the lower bound, $\widehat{a}_i/a_i \ge 1/c$ implies $\sum_i \widehat{a}_i^2/a_i \ge (1/c)\widehat{S}$, hence $\widehat{V} \ge \frac{1}{M^2}\frac{1/c}{\widehat{S}}$. Since $\widehat{a}_i \le c a_i$ implies $\widehat{S} \le cS$, we conclude $\widehat{V} \ge \frac{1}{M^2}\frac{1/c}{cS} = \frac{1}{c^2}V^\star$. □

Theorem 3.8 (Posterior-precision aggregation and scale cancellation)**.** *Consider the scalarized Gaussian model equation 15–equation 16 and assume the heteroscedastic variances factorize as*

$$v_i(\vartheta) = \sigma^2\, f_i(\vartheta), \qquad f_i(\vartheta) > 0, \tag{25}$$

so that the conditional precisions satisfy $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, f_i(\vartheta)^{-1}$. Fix ϑ (e.g., $\vartheta = (\beta, \xi)$) and endow σ^2 with any prior such that $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}_{j=1}^G]$ exists and is finite.

Define the posterior-precision scores

$$\widetilde{w}_i(\vartheta) \;:=\; \mathbb{E}\Big[\lambda_i(\vartheta)\,\Big|\,\vartheta, \{g_j\}_{j=1}^G\Big] \;=\; f_i(\vartheta)^{-1}\,\mathbb{E}\Big[\sigma^{-2}\,\Big|\,\vartheta, \{g_j\}_{j=1}^G\Big]. \tag{26}$$

Then the (unique) minimum-variance unbiased linear estimator of μ/M of the form $\widehat{\mu} = \sum_{i=1}^G x_i g_i$ under the constraint $\sum_{i=1}^G x_i = 1/M$ is obtained by normalizing the relative precisions:

$$x_i^{\text{Bayes}}(\vartheta) \;=\; \frac{1}{M}\,\frac{f_i(\vartheta)^{-1}}{\sum_{j=1}^G f_j(\vartheta)^{-1}} \;=\; \frac{1}{M}\,\frac{\widetilde{w}_i(\vartheta)}{\sum_{j=1}^G \widetilde{w}_j(\vartheta)}. \tag{27}$$

In particular, the common scale factor $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$ cancels from the normalized weights.

Proof. Fix ϑ and condition on (σ^2, ϑ). Under equation 25, the conditional variances are $v_i(\vartheta) = \sigma^2 f_i(\vartheta)$ and hence

$$v_i(\vartheta)^{-1} = \lambda_i(\vartheta) = \frac{1}{\sigma^2}\, f_i(\vartheta)^{-1}.$$

Therefore, conditional on (σ^2, ϑ) the variances $\{v_i(\vartheta)\}$ are known and Lemma 3.6 applies, yielding the unique minimizer

$$x_i^\star(\sigma^2, \vartheta) = \frac{1}{M}\,\frac{v_i(\vartheta)^{-1}}{\sum_{j=1}^G v_j(\vartheta)^{-1}} = \frac{1}{M}\,\frac{f_i(\vartheta)^{-1}}{\sum_{j=1}^G f_j(\vartheta)^{-1}},$$

where the factor σ^{-2} cancels from numerator and denominator. This is exactly the first equality in equation 27.

For the second equality, use equation 26 to write $\widetilde{w}_i(\vartheta) = f_i(\vartheta)^{-1}\,\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$; normalizing $\widetilde{w}_i(\vartheta)$ across i cancels the common factor $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$, giving the same weights as above. □

Remark 3.9 (When ϑ is random)*.* If ϑ itself is endowed with a non-degenerate prior (e.g., a prior on the length exponent β), then two distinct Bayesian constructions become relevant:

1. The *exact* posterior mean of the conditional MVU weights $x_i^\star(\vartheta) \propto f_i(\vartheta)^{-1}$, namely $\mathbb{E}[x_i^\star(\vartheta) \mid \{g_j\}]$.
2. The *posterior-precision approximation* that replaces $f_i(\vartheta)^{-1}$ by its posterior mean $\mathbb{E}[f_i(\vartheta)^{-1} \mid \{g_j\}]$ and then renormalizes across i.

These coincide whenever the posterior over ϑ is sufficiently concentrated or when $f_i(\vartheta)$ is treated as deterministic via Empirical Bayes (as done in §3.8).

Theorem 3.10 (Jeffreys' baseline recovers ΔL with $\alpha = 1$)**.** *Assume independence $s(\cdot;\xi) \equiv 1$, fix $\beta = 1$, and take $a_0, b_0 \to 0$ so that $p(\sigma^2) \propto 1/\sigma^2$. Then*

$$x_i^{\text{Bayes}} \;\propto\; L_i^{-1}. \qquad \text{(12 revisited)}$$

Proof. With $\beta = 1$ and $s \equiv 1$, we have $\lambda_i(\vartheta) = \sigma^{-2} L_i^{-1}$. From equation 22, $\mathbb{E}[\sigma^{-2} \mid \beta{=}1, \{g\}] = (G-1)/\mathrm{RSS}_\omega(\beta{=}1)$, a common factor that cancels in Theorem 3.8, yielding $x_i \propto L_i^{-1}$. □

Proposition 3.11 (Correctness-linked contraction implies $\alpha < 1$)**.** *Suppose the per-token variance contracts with response length via $\kappa(L) \in (0,1]$ with $\kappa'(L) \le 0$ and $\kappa(L) = c\,L^{-\delta}(1 + o(1))$ for some $\delta \in [0,1)$, so that*

$$\mathrm{Var}(g_i \mid L_i = L) = \sigma^2\, L\, \kappa(L)\, s(L;\xi) \;=\; \sigma^2\, L^{1-\delta}\, s(L;\xi)\,(1 + o(1)). \tag{28}$$

Then in the independence benchmark $s \equiv 1$ the implied ΔL exponent is $\alpha = \beta = 1 - \delta < 1$.

Proof. The effective length law in equation 28 gives variance exponent $\beta = 1 - \delta$. Applying Theorem 3.8 with Jeffreys' scale (or any scale prior whose common factor cancels) yields weights $\propto L^{-\beta}$, hence $\alpha = \beta = 1 - \delta < 1$. □

Corollary 3.12 (Serial dependence corrections)**.** *If $s(L;\xi) = \Theta(1)$ (e.g., AR(1) or k-band dependence), then the large-L exponent in Proposition 3.11 is unchanged; dependence enters through the multiplicative factor $s(L;\xi)^{-1}$ in the weights.*

3.4 Informative Prior on the Length Exponent

Retain independence ($s \equiv 1$) but let $\beta \sim \pi_\beta$ with posterior $p(\beta \mid \{g_i\})$. Using equation 27,

$$\widetilde{w}_i \;\propto\; \mathbb{E}\Big[L_i^{-\beta} \,\Big|\, \{g_j\}\Big]. \tag{29}$$

Two useful consequences:

1. *MAP/Laplace approximation.* If $p(\beta \mid \{g_i\})$ is sharply peaked at $\hat\beta$, then $x_i^{\text{Bayes}} \propto L_i^{-\hat\beta}$.
2. *Gaussian posterior moment.* If $\beta \mid \{g_i\} \sim \mathcal{N}(m_\beta, \tau_\beta^2)$ (ignoring truncation),

$$\mathbb{E}\Big[L_i^{-\beta} \mid \{g_i\}\Big] \;=\; \exp\Big(-m_\beta \ln L_i + \tfrac{1}{2}\tau_\beta^2 (\ln L_i)^2\Big) \;\approx\; L_i^{-m_\beta} \quad (\tau_\beta^2 \text{ small}), \tag{30}$$

   i.e., an effective $\alpha \simeq m_\beta$.

3.5 Why $\beta < 1$ When Longer Responses Are More Likely Correct

Intuition. If longer responses have a higher probability of being correct, then, after baselining, their token contributions are less noisy on average. In a heteroscedastic model, the per-token variance decreases with length, so the sum over L tokens grows sub-linearly in L. Consequently, inverse-variance weights scale as $L^{-\alpha}$ with $\alpha < 1$.

Heteroskedastic variance model tied to correctness. Let $g_i = \sum_{t=1}^{L_i} Z_{i,t}$ be the scalarized trajectory gradient (see equation 5 and equation 16). Suppose

$$\mathrm{Var}(Z_{i,t} \mid L_i = L) \;=\; \sigma^2\,\kappa(L), \qquad \kappa'(L) \le 0, \tag{31}$$

with $\kappa(L) \propto p(L)\{1 - p(L)\}$ for a correctness probability $p(L)$ with $p(L) \uparrow 1$ as $L \uparrow \infty$. Then

$$\mathrm{Var}(g_i \mid L_i = L) \;=\; \sigma^2\,L\,\kappa(L)\,s(L;\xi). \tag{32}$$

If $\kappa(L) = c\,L^{-\delta}(1 + o(1))$ with $\delta \in [0, 1)$, we recover equation 28 and thus $\alpha = 1 - \delta < 1$ by Proposition 3.11. The conclusion persists with serial dependence by Corollary 3.12.

3.6 Weights under k-banded Covariance Structure

We work with the scalarized series $Y_{i,t} := \boldsymbol{u}^\top \boldsymbol{Z}_{i,t}$ and $g_i = \sum_{t=1}^{L_i} Y_{i,t}$ as in equation 15. Assume second–order stationarity and the *k-banded, stretched-geometric* auto-covariance:

$$\gamma(0) = \mathrm{Var}(Y_{i,t}) = \tau^2, \qquad \gamma(h) = \mathrm{Cov}(Y_{i,t}, Y_{i,t-h}) = \tau^2\,\rho^{|h|^\eta}\;\mathbf{1}\{|h| \le k\}, \quad \eta \ge 0, \tag{33}$$

with $|\rho| < 1$ for $\eta > 0$. The case $\eta = 1$ reduces to geometric decay $\gamma(h) = \tau^2\rho^{|h|}$; $\eta \downarrow 0$ yields *no decay across lags* $\gamma(h) = \tau^2\rho$ (compound-symmetry within the band). Let $m_i := \min\{k, L_i - 1\}$.

Lemma 3.13 (Dependence shape factor for stretched-geometric k-band)**.** *Under equation 33,*

$$\mathrm{Var}(g_i) \;=\; \tau^2\left[L_i \;+\; 2\sum_{h=1}^{m_i}(L_i - h)\,\rho^{h^\eta}\right] \;=\; \tau^2\,L_i\,s(L_i;\rho,k,\eta), \qquad s(L;\rho,k,\eta) \;=\; 1 + \frac{2}{L}\sum_{h=1}^{m}(L-h)\rho^{h^\eta}. \tag{34}$$

Proof. By Bartlett's identity (Lemma C.1), $\mathrm{Var}(g_i) = \sum_{h=-(L_i-1)}^{L_i-1}(L_i - |h|)\gamma(h)$. Insert equation 33, collect positive and negative lags, and truncate at m_i. □

Proposition 3.14 (Weighting rule under stretched-geometric k-band)**.** *Under the heteroscedastic model equation 16 with $v_i(\vartheta) = \sigma^2 L_i^\beta\,s(L_i;\rho,k,\eta)$, the posterior-precision (Bayes) weights satisfy, up to a common factor,*

$$x_i^{\mathrm{Bayes}} \;\propto\; \frac{1}{L_i^\beta\,s(L_i;\rho,k,\eta)} \;=\; \frac{1}{L_i^\beta\left(1 + \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i - h)\rho^{h^\eta}\right)}. \tag{35}$$

In particular, under the Jeffreys baseline $\beta = 1$, $x_i \propto \left[L_i + 2\sum_{h=1}^{m_i}(L_i - h)\rho^{h^\eta}\right]^{-1}$.

Proof. From equation 16 and Lemma 3.13, $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, L_i^{-\beta}\, s(L_i;\rho,k,\eta)^{-1}$. Apply Theorem 3.8 and note $1/\sigma^2$ cancels upon normalization. □

Lemma 3.15 (Monotonicity in the decay exponent)**.** *For fixed (ρ, k, L) with $\rho \in (0,1)$, $s(L;\rho,k,\eta)$ is nonincreasing in η:*

$$0 \le \eta_1 < \eta_2 \;\Rightarrow\; s(L;\rho,k,\eta_1) \;\ge\; s(L;\rho,k,\eta_2).$$

Consequently, the weight magnitude $[L^\beta s(L;\rho,k,\eta)]^{-1}$ is nondecreasing in η.

Proof. For $h \ge 1$ and $\rho \in (0,1)$, the map $\eta \mapsto \rho^{h^\eta}$ is decreasing, hence each summand in equation 34 decreases in η. □

No-decay limit ($\eta = 0$): constant-lag covariance within the band. Setting $\eta = 0$ in equation 33 gives $\gamma(h) = \tau^2\rho$ for $1 \le |h| \le k$. Then

$$s(L;\rho,k,0) \;=\; 1 + \frac{2\rho}{L}\sum_{h=1}^{m}(L-h) \;=\; 1 + 2m\rho - \frac{m(m+1)}{L}\rho, \qquad m = \min\{k, L-1\}. \tag{36}$$

Accordingly,

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\Big(1 + 2m_i\rho - \frac{m_i(m_i+1)}{L_i}\rho\Big)}, \qquad m_i = \min\{k, L_i - 1\}. \tag{37}$$

Compound symmetry. Taking $k \ge L-1$ in equation 36 yields $m = L-1$ and

$$s(L;\rho,k \ge L-1,0) = 1 + \rho(L-1) = (1-\rho) + \rho L, \tag{38}$$

which is the compound-symmetry shape from Lemma 3.33. Thus

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\,\big(1 + \rho(L_i - 1)\big)} \qquad \text{and, for } \beta = 1, \qquad x_i \;\propto\; \frac{1}{(1-\rho)L_i + \rho L_i^2}. \tag{39}$$

Remark 3.16 (Asymptotics). For fixed k and $L \to \infty$, $s(L;\rho,k,\eta) = 1 + 2\sum_{h=1}^{k}\rho^{h^\eta} + O(L^{-1})$, a positive constant depending on (ρ,k,η). Hence the length exponent is unchanged and dependence multiplies the ΔL rule by a constant. Under compound symmetry (no band), $s(L;\rho,\,\cdot,0) = (1-\rho) + \rho L$, so $v_i = \Theta(L_i^{\beta+1})$ and, under $\beta = 1$, $x_i \sim (\rho L_i^2)^{-1}$ as $L_i \to \infty$.

3.7 Why $\beta > 1$ When Longer Responses Are More Likely Correct (with tail degradation)

Intuition. If longer responses are more likely to be correct up to a point, then after baselining, their token contributions become less noisy as length grows toward an optimum length $L^\star$. Beyond $L^\star$, if the model gets lost, correctness drops and noise increases again. In a heteroskedastic model, this corresponds to a *U-shaped* per-token variance profile $\kappa(L)$: initially decreasing, then increasing. Consequently, the variance of the sum over L tokens grows *sublinearly* before $L^\star$ and *superlinearly* in the tail, yielding inverse-variance weights that scale like $L^{-\beta(L)}$ with $\beta(L) < 1$ for moderate L and $\beta(L) > 1$ in the tail.

Correctness-weighted Heteroscedastic variance model (non-monotone variance profile). Let $g_i = \sum_{t=1}^{L_i} Z_{i,t}$ be the scalarized trajectory gradient (see equation 5 and equation 16). As before, write

$$\mathrm{Var}(Z_{i,t} \mid L_i = L) \;=\; \sigma^2\,\kappa(L), \qquad \mathrm{Var}(g_i \mid L_i = L) \;=\; \sigma^2\,L\,\kappa(L)\,s(L;\xi), \tag{40}$$

where $s(L;\xi)$ encodes serial dependence (Sec. 3.6). We now allow $\kappa(L)$ to be *non-monotone*:

Assumption 3.17 (Tail degradation)**.** *There exist $L^\star \in (1,\infty)$ and exponents $\delta \in [0,1)$, $\theta > 0$, and constants $c_-, c_+ > 0$ such that*

$$\kappa(L) = \begin{cases} c_-\,L^{-\delta}\big(1+o(1)\big), & L \nearrow L^\star \text{ (moderate regime)},\\ c_+\,L^{\;\theta}\big(1+o(1)\big), & L \to \infty \text{ (tail regime)}. \end{cases}$$

That is, κ decreases polynomially up to $L^\star$ and increases polynomially beyond it.

Local length exponent and effective law. Define the *local variance elasticity*

$$\varepsilon(L) \;:=\; \frac{d\log\kappa(L)}{d\log L}.$$

From equation 40, $\log v(L) = \log\sigma^2 + \log L + \log\kappa(L) + \log s(L;\xi)$, so (ignoring the slowly varying $s(L;\xi)$ for the exponent),

$$\beta_{\mathrm{eff}}(L) \;:=\; \frac{d\log v(L)}{d\log L} \;\approx\; 1+\varepsilon(L). \tag{41}$$

Assumption 3.17 implies

$$\beta_{\mathrm{eff}}(L) \approx \begin{cases} 1-\delta, & L \lesssim L^\star,\\ 1+\theta, & L \gg L^\star, \end{cases} \qquad \text{so} \quad v(L) \;\asymp\; \begin{cases} \sigma^2\,L^{\,1-\delta}\,s(L;\xi), & L \lesssim L^\star,\\ \sigma^2\,L^{\,1+\theta}\,s(L;\xi), & L \gg L^\star. \end{cases}$$

Proposition 3.18 (Posterior-precision weights with tail degradation)**.** *Under the heteroskedastic Normal model with variance $v(L)$ as in equation 40 and Assumption 3.17, the Bayes (posterior-precision) weights satisfy, up to a common factor,*

$$x_i^{\mathrm{Bayes}} \;\propto\; \frac{1}{v(L_i)} \;\approx\; \begin{cases} L_i^{-(1-\delta)}\,s(L_i;\xi)^{-1}, & L_i \lesssim L^\star,\\ L_i^{-(1+\theta)}\,s(L_i;\xi)^{-1}, & L_i \gg L^\star. \end{cases} \tag{42}$$

Equivalently, the implied ΔL exponent is $\beta(L) \approx \beta_{\mathrm{eff}}(L)$, i.e., $\beta(L) < 1$ before $L^\star$ and $\beta(L) > 1$ in the tail.

Proof. Theorem 3.8 yields $x_i^{\mathrm{Bayes}} \propto \mathbb{E}[1/v(L_i) \mid \text{data}]$. Under Assumption 3.17, $v(L)$ is regularly varying with index $1-\delta$ for $L \lesssim L^\star$ and $1+\theta$ for $L \to \infty$, hence the stated asymptotics. The multiplicative $s(L;\xi)$ enters as in Sec. 3.6. □

Impact of serial dependence. The factor $s(L;\xi)$ multiplies $v(L)$ and hence divides the weights:

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i\,\kappa(L_i)\,s(L_i;\xi)}.$$

For k-banded dependence with fixed k (Sec. 3.6), $s(L;\xi) \to s_\infty(\xi)$, a constant; the exponents in equation 3.7 remain $1 \pm (\delta, \theta)$. Under compound symmetry (constant-lag covariance; Sec. 3.9), $s(L;\rho) = (1-\rho) + \rho L$ contributes an extra factor $\sim \rho L$ as $L \to \infty$, so the tail exponent becomes $2+\theta$ and

$$x_i^{\text{Bayes}} \;\propto\; L_i^{-(2+\theta)} \quad \text{(tail, compound symmetry)}.$$

Concrete parametric example (unimodal correctness). Let the correctness probability be unimodal in $\log L$, e.g.,

$$p(L) = \Big(1 + \exp\{-a_0 - a_1 \log L + a_2 (\log L)^2\}\Big)^{-1}, \qquad a_2 > 0,$$

so $p(L) \uparrow$ for small L and $p(L) \downarrow$ for large L. If the variance factor is $\kappa(L) \propto p(L)\{1 - p(L)\}$ around the interior (maximal near $p = 1/2$) but is *augmented* by a tail noise term $\nu(L) \asymp L^\theta$ (credit-assignment drift), i.e.,

$$\kappa(L) \;\asymp\; c_- L^{-\delta} \;\text{ for } L \lesssim L^\star, \qquad \kappa(L) \;\asymp\; c_+ L^{\theta} \;\text{ for } L \gg L^\star,$$

then Proposition 3.18 applies with the same exponents.

Remark 3.19. Practical summary.

- For **moderate lengths** where correctness improves, use $\beta \simeq 1 - \delta < 1$.
- In the **long-tail** where the model gets lost, use $\beta \simeq 1 + \theta > 1$ (and add $+1$ more under compound symmetry).
- The posterior-precision rule automatically enacts this via $x_i \propto [L_i\,\kappa(L_i)\,s(L_i;\xi)]^{-1}$; the common scale cancels.

3.8 Empirical Bayes estimator for length and dependence

We consider trajectories $i = 1, \ldots, G$ with length $L_i \in \mathbb{N}$ and scalarized return $g_i \in \mathbb{R}$. Adopt the Normal working model

$$g_i \mid \mu, \sigma^2, \vartheta \sim \mathcal{N}\Big(\mu,\; v_i(\vartheta)\Big), \qquad v_i(\vartheta) = \sigma^2\, L_i^{\beta}\, s(L_i;\xi), \tag{43}$$

where $\mu \in \mathbb{R}$ is the common location, $\sigma^2 > 0$ is a global scale, $\beta \in \mathbb{R}$ (the length exponent) controls the length–variance law, and $s(L;\xi) > 0$ captures within-trajectory dependence via a low-dimensional shape ξ. We denote $\vartheta = (\beta, \xi)$. Define the precisions and normalized weights

$$\omega_i(\beta,\xi) = \frac{1}{L_i^{\beta}\, s(L_i;\xi)}, \qquad w_i(\beta,\xi) = \frac{\omega_i(\beta,\xi)}{\sum_{j=1}^{G} \omega_j(\beta,\xi)}. \tag{44}$$

We take $p(\mu) \propto 1$ and Jeffreys $p(\sigma^2) \propto 1/\sigma^2$; weak priors π_β, π_ξ may be used or set flat for EB/type-II ML.

Weighted aggregator. The Bayes estimator of μ under equation 43 (for fixed ϑ) is the precision-weighted mean

$$m(\beta,\xi) = \sum_{i=1}^{G} w_i(\beta,\xi)\, g_i = \frac{\sum_{i=1}^{G} \omega_i(\beta,\xi)\, g_i}{\sum_{i=1}^{G} \omega_i(\beta,\xi)}. \tag{45}$$

Let

$$e_i(\beta,\xi) = g_i - m(\beta,\xi), \qquad \Lambda_\omega(\beta,\xi) = \sum_{i=1}^{G} \omega_i(\beta,\xi),$$

and

$$\mathrm{RSS}_\omega(\beta,\xi) = \sum_{i=1}^{G} \omega_i(\beta,\xi)\, e_i(\beta,\xi)^2. \tag{46}$$

Proposition 3.20 (Posterior in closed form, conditional on (β,ξ))**.** *Under equation 43 with $p(\mu) \propto 1$ and $p(\sigma^2) \propto \frac{1}{\sigma^2}$,*

$$\mu \mid \sigma^2, \mathbf{g}, \vartheta \sim \mathcal{N}\Big(m(\beta,\xi),\, \sigma^2/\Lambda_\omega(\beta,\xi)\Big), \quad \sigma^2 \mid \mathbf{g}, \vartheta \sim \mathrm{Inv-Gamma}\left(\frac{G-1}{2},\, \frac{\mathrm{RSS}_\omega(\beta,\xi)}{2}\right),$$

where $\mathbf{g} = (g_1, \ldots, g_G)$ and we use the inverse-gamma parameterization with density $p(x) \propto x^{-(a+1)} \exp(-b/x)$ for $x > 0$, shape $a > 0$ and scale $b > 0$.

Proof. We first write down the likelihood explicitly. Given $\vartheta = (\beta,\xi)$,

$$v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i;\xi) = \frac{\sigma^2}{\omega_i(\beta,\xi)}.$$

Thus the conditional density of g_i is

$$p(g_i \mid \mu, \sigma^2, \vartheta) = \frac{1}{\sqrt{2\pi v_i(\vartheta)}} \exp\left(-\frac{(g_i-\mu)^2}{2v_i(\vartheta)}\right) = \frac{\sqrt{\omega_i}}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\omega_i}{2\sigma^2}(g_i-\mu)^2\right),$$

where, for brevity, we write $\omega_i = \omega_i(\beta,\xi)$. By independence across i,

$$p(\mathbf{g} \mid \mu, \sigma^2, \vartheta) = \prod_{i=1}^{G} p(g_i \mid \mu, \sigma^2, \vartheta) = (2\pi\sigma^2)^{-G/2} \left(\prod_{i=1}^{G} \omega_i^{1/2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i-\mu)^2\right).$$

Multiplying by the priors $p(\mu) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$ yields the unnormalized joint posterior

$$p(\mu, \sigma^2 \mid \mathbf{g}, \vartheta) \;\propto\; (\sigma^2)^{-G/2} \frac{1}{\sigma^2} \left(\prod_{i=1}^{G} \omega_i^{1/2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i-\mu)^2\right). \tag{47}$$

Next we complete the square in μ. Let

$$T = \sum_{i=1}^{G} \omega_i g_i, \qquad \Lambda_\omega = \sum_{i=1}^{G} \omega_i, \qquad m = \frac{T}{\Lambda_\omega}.$$

Then

$$\sum_{i=1}^{G}\omega_i(g_i-\mu)^2=\sum_{i=1}^{G}\omega_i\Big[(g_i-m)+(m-\mu)\Big]^2=\sum_{i=1}^{G}\omega_i(g_i-m)^2+2(m-\mu)\sum_{i=1}^{G}\omega_i(g_i-m)+\Lambda_\omega(\mu-m)^2.$$

The middle term vanishes because

$$\sum_{i=1}^{G}\omega_i(g_i-m)=\sum_{i=1}^{G}\omega_i g_i-m\sum_{i=1}^{G}\omega_i=T-\Lambda_\omega m=0.$$

Hence

$$\sum_{i=1}^{G}\omega_i(g_i-\mu)^2=\Lambda_\omega(\mu-m)^2+\sum_{i=1}^{G}\omega_i(g_i-m)^2=\Lambda_\omega(\mu-m)^2+\mathrm{RSS}_\omega,$$

with RSS_ω as in equation 46. Substituting into equation 47 gives

$$p(\mu,\sigma^2\mid\mathbf{g},\vartheta)\ \propto\ (\sigma^2)^{-(G/2+1)}\left(\prod_{i=1}^{G}\omega_i^{1/2}\right)\exp\left(-\frac{\Lambda_\omega}{2\sigma^2}(\mu-m)^2\right)\exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2}\right).$$

Conditional posterior of μ. For fixed σ^2, the dependence on μ is exactly that of a Normal kernel with mean m and variance σ^2/Λ_ω. Normalizing in μ therefore yields

$$\mu\mid\sigma^2,\mathbf{g},\vartheta\sim\mathcal{N}\Big(m(\beta,\xi),\ \sigma^2/\Lambda_\omega(\beta,\xi)\Big).$$

Marginal posterior of σ^2. Integrate out μ from $p(\mu,\sigma^2\mid\mathbf{g},\vartheta)$. The μ-integral is

$$\int_{-\infty}^{\infty}\exp\left(-\frac{\Lambda_\omega}{2\sigma^2}(\mu-m)^2\right)\,d\mu=\sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}}.$$

Thus

$$p(\sigma^2\mid\mathbf{g},\vartheta)\ \propto\ (\sigma^2)^{-(G/2+1)}\sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}}\left(\prod_{i=1}^{G}\omega_i^{1/2}\right)\exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2}\right).$$

The multiplicative constant $\sqrt{2\pi/\Lambda_\omega}\prod_i\omega_i^{1/2}$ does not depend on σ^2, so we drop it. We are left with

$$p(\sigma^2\mid\mathbf{g},\vartheta)\ \propto\ (\sigma^2)^{-(G/2+1-1/2)}\exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2}\right)=(\sigma^2)^{-(G+1)/2}\exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2}\right).$$

Comparing with the inverse-gamma kernel $x^{-(a+1)}\exp(-b/x)$ on $x>0$, we read off $a=(G-1)/2$ and $b=\mathrm{RSS}_\omega/2$. This proves the claimed inverse-gamma posterior for σ^2. □

Proposition 3.21 (GLS optimality of the aggregator)**.** *Fix (β,ξ). Among all linear unbiased estimators $\tilde\mu=\sum_{i=1}^{G}a_ig_i$ with $\sum_{i=1}^{G}a_i=1$, $m(\beta,\xi)$ in equation 45 attains the minimum variance* $\mathrm{Var}(\tilde\mu)=\sigma^2/\Lambda_\omega(\beta,\xi)$.

Proof. For fixed ϑ, the model equation 43 implies

$$\mathbb{E}[g_i \mid \mu, \vartheta] = \mu, \qquad \mathrm{Var}(g_i \mid \mu, \vartheta) = v_i(\vartheta) = \frac{\sigma^2}{\omega_i},$$

and the g_i are conditionally independent. Consider a generic linear estimator $\tilde{\mu} = \sum_i a_i g_i$ with coefficients $a_i \in \mathbb{R}$.

Unbiasedness constraint.

$$\mathbb{E}[\tilde{\mu} \mid \mu, \vartheta] = \sum_{i=1}^{G} a_i \mathbb{E}[g_i \mid \mu, \vartheta] = \sum_{i=1}^{G} a_i \mu = \mu \sum_{i=1}^{G} a_i.$$

Thus $\tilde{\mu}$ is unbiased for μ if and only if $\sum_{i=1}^{G} a_i = 1$.

Variance under the constraint. Using independence and $\mathrm{Var}(g_i) = \sigma^2/\omega_i$,

$$\mathrm{Var}(\tilde{\mu} \mid \mu, \vartheta) = \sum_{i=1}^{G} a_i^2 \,\mathrm{Var}(g_i \mid \mu, \vartheta) = \sigma^2 \sum_{i=1}^{G} \frac{a_i^2}{\omega_i}.$$

We must minimize $\sum_i a_i^2/\omega_i$ subject to $\sum_i a_i = 1$. Introduce a Lagrange multiplier λ and consider

$$\mathcal{L}(a_1, \ldots, a_G, \lambda) = \sum_{i=1}^{G} \frac{a_i^2}{\omega_i} - \lambda \left(\sum_{i=1}^{G} a_i - 1 \right).$$

Setting partial derivatives to zero,

$$\frac{\partial \mathcal{L}}{\partial a_i} = \frac{2a_i}{\omega_i} - \lambda = 0 \quad \Rightarrow \quad a_i = \frac{\lambda}{2}\omega_i, \qquad i = 1, \ldots, G.$$

Imposing $\sum_i a_i = 1$ gives

$$\sum_{i=1}^{G} a_i = \frac{\lambda}{2} \sum_{i=1}^{G} \omega_i = \frac{\lambda}{2} \Lambda_\omega = 1 \quad \Rightarrow \quad \lambda = \frac{2}{\Lambda_\omega}.$$

Thus

$$a_i = \frac{\omega_i}{\Lambda_\omega} = w_i(\beta, \xi),$$

and the unique minimizer is exactly the weight vector in equation 45. The corresponding variance is

$$\mathrm{Var}(m(\beta, \xi) \mid \mu, \vartheta) = \sigma^2 \sum_{i=1}^{G} \frac{w_i^2}{\omega_i} = \sigma^2 \sum_{i=1}^{G} \frac{\omega_i^2}{\Lambda_\omega^2 \, \omega_i} = \sigma^2 \frac{\sum_i \omega_i}{\Lambda_\omega^2} = \frac{\sigma^2}{\Lambda_\omega}.$$

Hence $m(\beta, \xi)$ is the Gauss–Markov / GLS estimator with minimum variance among all linear unbiased estimators. □

Theorem 3.22 (EB marginal objective for (β, ξ))**.** *Under equation 43 with the above priors, the marginal type-II log-posterior for $\vartheta = (\beta, \xi)$ is (up to an additive constant)*

$$\ell(\beta, \xi) = \log \pi_\beta(\beta) + \log \pi_\xi(\xi) + \tfrac{1}{2} \sum_{i=1}^{G} \log \omega_i(\beta, \xi) - \tfrac{1}{2} \log \Lambda_\omega(\beta, \xi) - \tfrac{G-1}{2} \log \mathrm{RSS}_\omega(\beta, \xi). \quad (48)$$

Proof. We integrate out μ and σ^2 from the joint model

$$p(\mathbf{g}, \mu, \sigma^2 \mid \vartheta) = p(\mathbf{g} \mid \mu, \sigma^2, \vartheta)\, p(\mu)\, p(\sigma^2)$$

and then add the priors π_β, π_ξ on (β, ξ).

From the likelihood calculation in the proof of Proposition 3.20, we have

$$p(\mathbf{g} \mid \mu, \sigma^2, \vartheta) = (2\pi\sigma^2)^{-G/2} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i - \mu)^2 \right),$$

and $\sum_{i=1}^{G} \omega_i (g_i - \mu)^2 = \Lambda_\omega (\mu - m)^2 + \mathrm{RSS}_\omega$. Using $p(\mu) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$,

$$p(\mathbf{g}, \mu, \sigma^2 \mid \vartheta) \propto (\sigma^2)^{-(G/2+1)} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right).$$

Integrate with respect to μ. As in Proposition 3.20,

$$\int_{-\infty}^{\infty} \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) d\mu = \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}},$$

hence

$$\begin{aligned} p(\mathbf{g}, \sigma^2 \mid \vartheta) &\propto (\sigma^2)^{-(G/2+1)} \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right) \\ &\propto (\sigma^2)^{-(G+1)/2} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \Lambda_\omega^{-1/2} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right), \end{aligned}$$

where we have absorbed constants depending on 2π into a factor that does not depend on ϑ.

Now integrate out σ^2 using the standard identity for inverse-gamma integrals:

$$\int_0^{\infty} (\sigma^2)^{-(a+1)} \exp\left(-\frac{b}{\sigma^2} \right) d\sigma^2 = 2\, b^{-a}\, \Gamma(a), \qquad a > 0,\ b > 0.$$

Here we take $a = (G-1)/2$, $b = \mathrm{RSS}_\omega/2$. Thus

$$\int_0^{\infty} (\sigma^2)^{-(G+1)/2} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right) d\sigma^2 \propto \mathrm{RSS}_\omega^{-(G-1)/2}.$$

It follows that the marginal likelihood of the data given ϑ is

$$p(\mathbf{g} \mid \vartheta) \propto \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \Lambda_\omega^{-1/2} \mathrm{RSS}_\omega^{-(G-1)/2},$$

up to factors independent of ϑ. Taking logarithms and adding $\log \pi_\beta(\beta) + \log \pi_\xi(\xi)$ yields

$$\ell(\beta, \xi) = \log p(\mathbf{g} \mid \vartheta) + \log \pi_\beta(\beta) + \log \pi_\xi(\xi) = \log \pi_\beta(\beta) + \log \pi_\xi(\xi) + \frac{1}{2} \sum_{i=1}^{G} \log \omega_i - \frac{1}{2} \log \Lambda_\omega - \frac{G-1}{2} \log \mathrm{RSS}_\omega + C$$

where C does not depend on (β, ξ). Dropping the additive constant proves equation 48. $\square$

Lemma 3.23 (Weighted RSS derivative identity)**.** *For any component $\phi \in \{\beta\} \cup \{\xi \text{ components}\}$,*

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega(\beta, \xi) = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 = \sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}. \tag{49}$$

Proof. Recall $\mathrm{RSS}_\omega(\beta, \xi) = \sum_{i=1}^{G} \omega_i e_i^2$, with $e_i = g_i - m$ and $m = m(\beta, \xi)$. We differentiate RSS_ω with respect to ϕ. Using the product and chain rules,

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \left[\frac{\partial \omega_i}{\partial \phi} e_i^2 + \omega_i \frac{\partial}{\partial \phi}(e_i^2) \right].$$

Since $e_i = g_i - m$ and g_i does not depend on ϕ, $\frac{\partial e_i}{\partial \phi} = -\frac{\partial m}{\partial \phi}$, so

$$\frac{\partial}{\partial \phi}(e_i^2) = 2e_i \frac{\partial e_i}{\partial \phi} = 2e_i \left(-\frac{\partial m}{\partial \phi} \right) = -2e_i \frac{\partial m}{\partial \phi}.$$

Therefore,

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 - 2 \frac{\partial m}{\partial \phi} \sum_{i=1}^{G} \omega_i e_i.$$

To simplify the second term, observe that

$$\sum_{i=1}^{G} \omega_i e_i = \sum_{i=1}^{G} \omega_i (g_i - m) = \sum_{i=1}^{G} \omega_i g_i - m \sum_{i=1}^{G} \omega_i = T - m\Lambda_\omega = 0,$$

because $m = T/\Lambda_\omega$. Thus the term involving $\partial m / \partial \phi$ cancels, and we obtain

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2.$$

Finally, use $\frac{\partial \omega_i}{\partial \phi} = \omega_i \frac{\partial \log \omega_i}{\partial \phi}$ to get the second equality:

$$\sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 = \sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}.$$

□

Proposition 3.24 (Gradient of the EB objective)**.** *With $\Lambda_\omega, \mathrm{RSS}_\omega$ as above,*

$$\frac{\partial \ell}{\partial \phi} = \frac{1}{2} \left[\sum_{i=1}^{G} \frac{\partial \log \omega_i}{\partial \phi} - \frac{1}{\Lambda_\omega} \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} \right] - \frac{G-1}{2} \frac{1}{\mathrm{RSS}_\omega} \sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi} + \frac{\partial \log \pi(\phi)}{\partial \phi}, \tag{50}$$

where $\pi(\phi)$ denotes the marginal prior factor in $\pi_\beta \pi_\xi$ involving ϕ.

Proof. Starting from equation 48, we differentiate term by term with respect to ϕ.

Prior term. By definition,

$$\frac{\partial}{\partial \phi} \Big(\log \pi_\beta(\beta) + \log \pi_\xi(\xi) \Big) = \frac{\partial \log \pi(\phi)}{\partial \phi},$$

where $\pi(\phi)$ denotes the appropriate prior factor.

Sum of log-precisions. For the third term in equation 48,

$$\frac{\partial}{\partial\phi}\left(\frac{1}{2}\sum_{i=1}^{G}\log\omega_i\right)=\frac{1}{2}\sum_{i=1}^{G}\frac{1}{\omega_i}\frac{\partial\omega_i}{\partial\phi}=\frac{1}{2}\sum_{i=1}^{G}\frac{\partial\log\omega_i}{\partial\phi}.$$

Log of total precision. For the fourth term,

$$\frac{\partial}{\partial\phi}\left(-\frac{1}{2}\log\Lambda_\omega\right)=-\frac{1}{2}\frac{1}{\Lambda_\omega}\frac{\partial\Lambda_\omega}{\partial\phi}=-\frac{1}{2}\frac{1}{\Lambda_\omega}\sum_{i=1}^{G}\frac{\partial\omega_i}{\partial\phi}.$$

Residual sum of squares term. For the final term,

$$\frac{\partial}{\partial\phi}\left(-\frac{G-1}{2}\log\mathrm{RSS}_\omega\right)=-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\frac{\partial\mathrm{RSS}_\omega}{\partial\phi}.$$

By Lemma 3.23, $\frac{\partial\mathrm{RSS}_\omega}{\partial\phi}=\sum_i\omega_i e_i^2\frac{\partial\log\omega_i}{\partial\phi}$, hence

$$-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\frac{\partial\mathrm{RSS}_\omega}{\partial\phi}=-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2\frac{\partial\log\omega_i}{\partial\phi}.$$

Collecting the contributions from all terms gives equation 50. □

Corollary 3.25 (Gradient in the length direction)**.** *Since* $\log\omega_i=-\beta\log L_i-\log s(L_i;\xi)$, $\partial\log\omega_i/\partial\beta=-\log L_i$ *and* $\partial\omega_i/\partial\beta=-\omega_i\log L_i$. *Hence*

$$\frac{\partial\ell}{\partial\beta}=\frac{1}{2}\left[-\sum_{i=1}^{G}\log L_i+\frac{\sum_{i=1}^{G}\omega_i\log L_i}{\Lambda_\omega}\right]+\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2\log L_i+\frac{\partial\log\pi_\beta(\beta)}{\partial\beta}. \tag{51}$$

Proof. Apply Proposition 3.24 with $\phi=\beta$. We have

$$\frac{\partial\log\omega_i}{\partial\beta}=-\log L_i,\qquad\frac{\partial\omega_i}{\partial\beta}=\omega_i\frac{\partial\log\omega_i}{\partial\beta}=-\omega_i\log L_i.$$

Substitute these into equation 50. The first bracketed term becomes

$$\begin{aligned}\frac{1}{2}\left[\sum_{i=1}^{G}\frac{\partial\log\omega_i}{\partial\beta}-\frac{1}{\Lambda_\omega}\sum_{i=1}^{G}\frac{\partial\omega_i}{\partial\beta}\right]&=\frac{1}{2}\left[\sum_{i=1}^{G}(-\log L_i)-\frac{1}{\Lambda_\omega}\sum_{i=1}^{G}(-\omega_i\log L_i)\right]\\&=\frac{1}{2}\left[-\sum_{i=1}^{G}\log L_i+\frac{\sum_{i=1}^{G}\omega_i\log L_i}{\Lambda_\omega}\right].\end{aligned}$$

The second term in equation 50 becomes

$$\begin{aligned}-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2\frac{\partial\log\omega_i}{\partial\beta}&=-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2(-\log L_i)\\&=\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2\log L_i.\end{aligned}$$

Adding the prior derivative $\partial\log\pi_\beta(\beta)/\partial\beta$ yields equation 51. □

Definition 3.26 (Stretched-geometric k-band dependence)**.** *Given $k \in \mathbb{N}$, $\rho \in (-1, 1)$, $\eta \geq 0$, define with $m = \min\{k, L-1\}$*

$$s(L; \rho, k, \eta) = 1 + \frac{2}{L} \sum_{h=1}^{m} (L-h)\, \rho^{h^\eta}. \tag{52}$$

Lemma 3.27 (Derivatives of the k-band shape)**.** *Let $s_i = s(L_i; \rho, k, \eta)$, $m_i = \min\{k, L_i - 1\}$. Then*

$$\frac{\partial \log \omega_i}{\partial \rho} = -\frac{1}{s_i} \frac{\partial s_i}{\partial \rho}, \quad \frac{\partial s_i}{\partial \rho} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, h^\eta\, \rho^{h^\eta - 1},$$

$$\frac{\partial \log \omega_i}{\partial \eta} = -\frac{1}{s_i} \frac{\partial s_i}{\partial \eta}, \quad \frac{\partial s_i}{\partial \eta} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho^{h^\eta} (\log \rho)\, h^\eta \log h.$$

Proof. By Definition 3.26,

$$s_i = 1 + \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho^{h^\eta}.$$

For $\partial s_i / \partial \rho$, note that $\partial(1)/\partial\rho = 0$ and we can differentiate the sum termwise:

$$\frac{\partial s_i}{\partial \rho} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \frac{\partial}{\partial \rho}\Big(\rho^{h^\eta}\Big).$$

Using $\rho^{h^\eta} = \exp(h^\eta \log \rho)$, the chain rule gives

$$\frac{\partial}{\partial \rho}\Big(\rho^{h^\eta}\Big) = h^\eta \rho^{h^\eta - 1},$$

so

$$\frac{\partial s_i}{\partial \rho} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, h^\eta\, \rho^{h^\eta - 1}.$$

For $\partial s_i / \partial \eta$, again differentiate termwise:

$$\frac{\partial s_i}{\partial \eta} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \frac{\partial}{\partial \eta}\Big(\rho^{h^\eta}\Big).$$

Write $\rho^{h^\eta} = \exp(h^\eta \log \rho)$. Then

$$\frac{\partial}{\partial \eta}\Big(\rho^{h^\eta}\Big) = \rho^{h^\eta} \frac{\partial}{\partial \eta}\Big(h^\eta \log \rho\Big) = \rho^{h^\eta} (\log \rho) h^\eta \log h.$$

Thus

$$\frac{\partial s_i}{\partial \eta} = \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho^{h^\eta} (\log \rho) h^\eta \log h.$$

Finally, since $\omega_i = L_i^{-\beta} s(L_i; \xi)^{-1}$,

$$\log \omega_i = -\beta \log L_i - \log s_i,$$

so

$$\frac{\partial \log \omega_i}{\partial \rho} = -\frac{1}{s_i} \frac{\partial s_i}{\partial \rho}, \qquad \frac{\partial \log \omega_i}{\partial \eta} = -\frac{1}{s_i} \frac{\partial s_i}{\partial \eta}.$$

□

Proposition 3.28 (Unbiasedness of $m(\beta, \xi)$)**.** *For fixed (β, ξ), the estimator $m(\beta, \xi)$ in equation 45 is unbiased for μ.*

Proof. By equation 43, for each i, $\mathbb{E}[g_i \mid \mu, \vartheta] = \mu$. Using equation 45,

$$\mathbb{E}[m(\beta, \xi) \mid \mu, \vartheta] = \mathbb{E}\left[\sum_{i=1}^{G} w_i g_i \,\middle|\, \mu, \vartheta\right] = \sum_{i=1}^{G} w_i \,\mathbb{E}[g_i \mid \mu, \vartheta] = \mu \sum_{i=1}^{G} w_i.$$

From equation 44, $\sum_i w_i = 1$. Therefore $\mathbb{E}[m(\beta, \xi) \mid \mu, \vartheta] = \mu$, i.e., $m(\beta, \xi)$ is unbiased. □

Proposition 3.29 (Link to the ΔL rule)**.** *If $s(L; \xi) \equiv 1$ and $\beta = 1$, then $\omega_i = 1/L_i$ and $w_i \propto 1/L_i$, recovering the ΔL baseline weighting.*

Proof. If $s(L; \xi) \equiv 1$ and $\beta = 1$, then for each i,

$$\omega_i = \frac{1}{L_i^{\beta} s(L_i; \xi)} = \frac{1}{L_i \cdot 1} = \frac{1}{L_i}.$$

Hence

$$w_i = \frac{\omega_i}{\sum_{j=1}^{G} \omega_j} = \frac{L_i^{-1}}{\sum_{j=1}^{G} L_j^{-1}} \propto \frac{1}{L_i},$$

which is the standard ΔL choice of weights. □

Assumption 3.30 (Regularity and identifiability)**.** *(i) $\{(g_i, L_i)\}_{i \geq 1}$ are i.i.d. or strictly stationary and ergodic with $\sup_i \mathbb{E}[g_i^4] < \infty$.*

(ii) The parameter space $\Theta = \{\beta\} \times \Xi$ is compact (for example, $\beta \in [\beta_{\min}, \beta_{\max}]$, $|\rho| \leq \rho_{\max} < 1$, $\eta \in [0, \eta_{\max}]$, and fixed k).

(iii) The population objective $\ell_\infty(\vartheta) := \lim_{G \to \infty} G^{-1} \mathbb{E}[\ell_G(\vartheta)]$ exists and has a unique maximizer $\vartheta^\star = (\beta^\star, \xi^\star)$.

(iv) $s(L; \xi)$ is smooth in ξ and bounded away from 0 and $+\infty$ on $\operatorname{supp}(L) \times \Xi$.

Here $\ell_G(\vartheta)$ denotes the EB objective equation 48 built from the first G trajectories; we may equivalently work with the rescaled version $G^{-1}\ell_G(\vartheta)$, which only differs by a factor independent of ϑ.

Theorem 3.31 (Consistency & asymptotic normality of the EB estimator)**.** *Let $\widehat{\vartheta}_G = \arg\max_{\vartheta \in \Theta} \ell_G(\vartheta)$ be any (measurable) maximizer. Under Assumption 3.30, $\widehat{\vartheta}_G \xrightarrow{p} \vartheta^\star$ as $G \to \infty$.*

If, in addition, ℓ_∞ is twice continuously differentiable at $\vartheta^\star$ with negative-definite Hessian $H = -\nabla^2 \ell_\infty(\vartheta^\star)$, and the gradient process satisfies a suitable central limit theorem (for example, $\sqrt{G}\, \nabla \ell_G(\vartheta^\star) \xrightarrow{d} \mathcal{N}(0, J)$ for some positive semi-definite matrix J), then

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) \;\xrightarrow{d}\; \mathcal{N}\Big(0,\; H^{-1} J H^{-1}\Big),$$

where J is the asymptotic covariance of $\sqrt{G}\, \nabla \ell_G(\vartheta^\star)$.

Proof. Consistency. We can write $\ell_G(\vartheta)$ as a continuous function of empirical averages of bounded (or suitably integrable) functions of (g_i, L_i). Indeed, the terms $\sum_i \log \omega_i$, Λ_ω, and RSS_ω are all sums of such functions, and the mapping from these empirical sums to $\ell_G(\vartheta)$ is continuous in both the empirical sums and in ϑ. By the (vector) law of large numbers and the continuous mapping theorem, for each fixed ϑ, the rescaled objective $G^{-1}\ell_G(\vartheta)$ converges almost surely to a deterministic limit $\ell_\infty(\vartheta)$. Under the compactness of Θ and smoothness/boundedness conditions in Assumption 3.30, standard results in M-estimation (e.g., an argmax continuous mapping theorem) imply that this convergence is uniform on Θ. By assumption, ℓ_∞ has a unique maximizer $\vartheta^\star$. The uniform convergence combined with uniqueness yields $\widehat{\vartheta}_G \xrightarrow{p} \vartheta^\star$ (see, e.g., van der Vaart (1998), Theorem 5.7).

Asymptotic normality. Consider the first-order optimality condition

$$\nabla \ell_G(\widehat{\vartheta}_G) = 0.$$

Assume ℓ_G is twice continuously differentiable in a neighborhood of $\vartheta^\star$ with probability tending to one, and expand $\nabla \ell_G$ in a Taylor series around $\vartheta^\star$:

$$\nabla \ell_G(\widehat{\vartheta}_G) = \nabla \ell_G(\vartheta^\star) + \nabla^2 \ell_G(\tilde{\vartheta}_G)(\widehat{\vartheta}_G - \vartheta^\star),$$

for some intermediate point $\tilde{\vartheta}_G$ on the line segment between $\widehat{\vartheta}_G$ and $\vartheta^\star$. Setting $\nabla \ell_G(\widehat{\vartheta}_G) = 0$ and rearranging gives

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) = -\Big(\nabla^2 \ell_G(\tilde{\vartheta}_G)\Big)^{-1} \sqrt{G}\, \nabla \ell_G(\vartheta^\star).$$

By the consistency already established, $\tilde{\vartheta}_G \xrightarrow{p} \vartheta^\star$. Under the stated differentiability and regularity assumptions, $\nabla^2 \ell_G(\tilde{\vartheta}_G)$ converges in probability to $\nabla^2 \ell_\infty(\vartheta^\star) = -H$, so $\Big(\nabla^2 \ell_G(\tilde{\vartheta}_G)\Big)^{-1} \xrightarrow{p} H^{-1}$.

By assumption, $\sqrt{G}\, \nabla \ell_G(\vartheta^\star) \xrightarrow{d} \mathcal{N}(0, J)$. Applying Slutsky's theorem to the product of the converging matrices and vectors yields

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) \xrightarrow{d} -H^{-1}\mathcal{N}(0, J) = \mathcal{N}(0, H^{-1} J H^{-1}),$$

which proves the stated asymptotic normality. □

Remark 3.32 (Smooth but not globally concave). While $\ell(\beta, \xi)$ is generally not globally concave, the gradient $\nabla \ell$ and local Hessian $\nabla^2 \ell$ are explicit via Lemma 3.23, Proposition 3.24, and Lemma 3.27. In practice, damping or exponential moving averages in (β, ξ) and simple box constraints on the parameter space Θ yield stable online updates; see the separate Algorithms section for concrete update rules.

3.9 Weights under non-vanishing constant-lag covariance (compound symmetry)

We work with the scalarized token series $Y_{i,t} := \boldsymbol{u}^\top \boldsymbol{Z}_{i,t}$ as in equation 15. Assume second–order stationarity and

$$\gamma(0) = \mathrm{Var}(Y_{i,t}) = \tau^2, \qquad \gamma(h) = \mathrm{Cov}(Y_{i,t}, Y_{i,t-h}) = \tau^2 \rho \quad \text{for all } h \neq 0, \tag{53}$$

with $\rho \in (-1/(L_i - 1),\, 1)$ to ensure positive definiteness for a given length L_i. (For uniformly valid models across unbounded L, take $\rho \in [0, 1)$.)

Lemma 3.33 (Trajectory variance under compound symmetry)**.** *Let $g_i = \sum_{t=1}^{L_i} Y_{i,t}$. Under equation 53,*

$$\mathrm{Var}(g_i) \;=\; \tau^2\left[L_i \;+\; 2\rho\sum_{h=1}^{L_i-1}(L_i - h)\right] \;=\; \tau^2\, L_i\Big(1+\rho(L_i-1)\Big) \;=\; \tau^2\Big((1-\rho)L_i+\rho L_i^2\Big). \tag{54}$$

Equivalently, with the shape factor $s(L;\rho) := \mathrm{Var}(g_i)/(\tau^2 L)$,

$$s(L;\rho) \;=\; 1+\rho(L-1) \;=\; (1-\rho)+\rho L. \tag{55}$$

Proof. By Bartlett's identity (Lemma C.1),

$$\mathrm{Var}(g_i) = \sum_{h=-(L_i-1)}^{L_i-1}(L_i - |h|)\gamma(h) = L_i\gamma(0) + 2\sum_{h=1}^{L_i-1}(L_i - h)\gamma(h).$$

Insert equation 53 to obtain the first equality in equation 54; the second follows from $\sum_{h=1}^{L-1}(L-h) = L(L-1)/2$. The shape factor is equation 55.

(Alternative matrix proof.) The $L_i \times L_i$ covariance matrix of $(Y_{i,1}, \ldots, Y_{i,L_i})$ is $\tau^2\Big((1-\rho)\boldsymbol{I} + \rho\,\mathbf{1}\mathbf{1}^\top\Big)$. Then $\mathrm{Var}(g_i) = \mathbf{1}^\top \mathrm{Cov}\,\mathbf{1} = \tau^2\Big((1-\rho)L_i + \rho L_i^2\Big)$. □

Proposition 3.34 (Weighting rule under constant-lag covariance)**.** *Under the heteroscedastic model equation 16 with $v_i(\vartheta) = \sigma^2 L_i^\beta\, s(L_i;\rho)$ and $s(L;\rho)$ from equation 55, the posterior-precision (Bayes) weights satisfy, up to a common factor,*

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\, s(L_i;\rho)} \;=\; \frac{1}{L_i^\beta\Big(1+\rho(L_i-1)\Big)}. \tag{56}$$

In particular, under the Jeffreys baseline with $\beta = 1$,

$$x_i \;\propto\; \frac{1}{L_i\Big(1+\rho(L_i-1)\Big)} \;=\; \frac{1}{(1-\rho)L_i+\rho L_i^2}. \tag{57}$$

Proof. From equation 16 and Lemma 3.33, $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, L_i^{-\beta}\, s(L_i;\rho)^{-1}$. By Theorem 3.8, $x_i^{\text{Bayes}} \propto \mathbb{E}[\lambda_i(\vartheta) \mid \{g\}]$; the factor $1/\sigma^2$ cancels upon normalization, yielding equation 56. The specialization $\beta = 1$ gives equation 57. □

Corollary 3.35 (Asymptotics and effective length law)**.** *If $\rho > 0$ is fixed and $L \to \infty$, then $s(L;\rho) \sim \rho L$ and*

$$v_i(\vartheta) \;\sim\; \sigma^2 L_i^\beta\, \rho L_i \quad\Longrightarrow\quad v_i(\vartheta) = \Theta(L_i^{\beta+1}).$$

Under the Jeffreys baseline $\beta = 1$, $v_i = \Theta(L_i^2)$ and $x_i \propto L_i^{-2}$ asymptotically. Thus persistent positive correlation induces a quadratic penalty in length relative to the i.i.d. case.

Remark 3.36 (Parameter constraints). For a fixed length L, compound symmetry is positive definite for $\rho \in (-1/(L-1), 1)$. To model batches with unbounded L, take $\rho \in [0, 1)$ so that $s(L;\rho)$ remains positive and the covariance is valid for all L.

4 Algorithms: CAPO with Empirical Bayes length and dependence estimation

Workflow overview (precise step-by-step)

We integrate Empirical Bayes (EB) estimation of the length exponent β and dependence shape ξ directly into the CAPO training loop, using the notation and objects introduced in §3.8 (in particular, the precisions $\omega_i(\beta, \xi)$, weights $w_i(\beta, \xi)$, aggregator $m(\beta, \xi)$, and summaries $\Lambda_\omega(\beta, \xi)$, $\mathrm{RSS}_\omega(\beta, \xi)$).

1. **Rollout collection.** At optimization step t, sample trajectories under the current policy π_{θ_t}. This yields a batch
$$\mathcal{B}_t = \Big\{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\Big\}_{i=1}^{G_t},$$
where $L_i \in \mathbb{N}$ is the trajectory length, $g_i \in \mathbb{R}$ is a scalarized trajectory return (e.g., episode reward or aggregated advantage), and $Y_{i,\tau} \in \mathbb{R}$ are per-token scalar increments used for estimating within-trajectory dependence.

2. **E-step statistics (given (β_{t-1}, ξ_{t-1})).** Using the current EB parameters (β_{t-1}, ξ_{t-1}), compute per-trajectory precisions and normalized weights as
$$\omega_i^{(t)} \;=\; \omega_i(\beta_{t-1}, \xi_{t-1}) \;=\; \Big(L_i^{\beta_{t-1}} s(L_i; \xi_{t-1})\Big)^{-1}, \qquad w_i^{(t)} \;=\; \frac{\omega_i^{(t)}}{\sum_{j\in\mathcal{B}_t} \omega_j^{(t)}}.$$
Form the batch aggregator (the empirical version of $m(\beta, \xi)$),
$$m_t \;=\; \sum_{i\in\mathcal{B}_t} w_i^{(t)} g_i,$$
the residuals $e_i^{(t)} = g_i - m_t$, and the summaries
$$\Lambda_\omega^{(t)} = \sum_{i\in\mathcal{B}_t} \omega_i^{(t)}, \qquad \mathrm{RSS}_\omega^{(t)} = \sum_{i\in\mathcal{B}_t} \omega_i^{(t)} \Big(e_i^{(t)}\Big)^2.$$

3. **M-step on EB objective.** Using the EB objective $\ell(\beta, \xi)$ from equation 48 and its gradients from Proposition 3.24 and Corollary 3.25, take damped/EMA-smoothed ascent steps on β and ξ. For ξ, either
   (i) use the *ACF-moment* estimator (Algorithm 2), which fits the dependence family $s(L; \rho, k, \eta)$ of Definition 3.26 to empirical autocorrelations, or
   (ii) directly use the EB gradient $\nabla_\xi \ell(\beta, \xi)$ in combination with the derivatives from Lemma 3.27.

4. **Reweight and update policy.** After updating to (β_t, ξ_t), recompute trajectory weights
$$w_i^{(t)} \;\propto\; \omega_i(\beta_t, \xi_t) \;=\; \Big(L_i^{\beta_t} s(L_i; \xi_t)\Big)^{-1},$$
and feed these into CAPO's policy/teacher update (e.g., reweighted SFT/OPD losses or variance-aware policy gradients).

5. **Stabilization & diagnostics.** Apply exponential moving averages (EMA) to (β, ξ), enforce simple box constraints such as $|\rho| \leq \rho_{\max} < 1$, $\eta \geq 0$, and clip extreme weights (e.g., the largest 0.5% of w_i). Monitor weight entropy and coefficient of variation as diagnostics, and update (β, ξ) every $2-4$ outer steps for additional stability.

Algorithm 1: EB–CAPO in the CAPO training loop

Algorithm 1 implements the full EB–CAPO loop described above. At each outer step t, it:

1. collects a batch $\mathcal{B}_t$ of trajectories $(g_i, L_i, \{Y_{i,\tau}\})$;
2. computes the E-step statistics under the current hyperparameters (β_{t-1}, ξ_{t-1}), namely the precisions $\omega_i^{(t)}$, normalized weights $w_i^{(t)}$, weighted mean m_t, residuals $e_i^{(t)}$, and summaries $\Lambda_\omega^{(t)}, \mathrm{RSS}_\omega^{(t)}$;
3. performs an M-step for the length exponent β using the EB gradient $\partial_\beta \ell(\beta, \xi)$ in Corollary 3.25, with EMA and box-constraint stabilization;
4. updates the dependence parameter ξ either via a plug-in ACF-moment fit (Algorithm 2) or via the EB gradient $\nabla_\xi \ell(\beta, \xi)$ from Proposition 3.24 and Lemma 3.27;
5. recomputes weights at the updated (β_t, ξ_t) and passes them to the CAPO policy/teacher loss.

Thus Algorithm 1 provides an online, amortized approximation to maximizing the EB objective $\ell(\beta, \xi)$ while simultaneously using the resulting precision weights in the CAPO optimizer.

Algorithm 2: ACF-moment estimator for dependence

Algorithm 2 provides a moment-based plug-in estimator for the dependence shape $\xi = (\rho, \eta, \dots)$ in the k-banded stretched-geometric family $s(L; \rho, k, \eta)$ (Definition 3.26). It operates at the *token* level $Y_{i,\tau}$ and:

1. computes per-trajectory empirical autocovariances $\hat{\gamma}_i(h)$ up to lag k, centered by the within-trajectory mean $\bar{Y}_i$;
2. forms a pooled autocovariance $\bar{\gamma}(h)$ by weighting each trajectory i by the number of contributing pairs $L_i - h$;
3. converts to a pooled autocorrelation $\bar{r}(h) = \bar{\gamma}(h)/\bar{\gamma}(0)$;
4. fits (ρ, η) by least squares against the theoretical correlation ρ^{h^η} implied by the stretched-geometric model.

The resulting $\tilde{\xi} = (\tilde{\rho}, \tilde{\eta}, \dots)$ can be used directly in the variance law $v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i; \xi)$, or as a warm start for subsequent EB gradient updates in Algorithm 1 or Algorithm 5.

Algorithm 1: EB–CAPO: Online EB estimation and trajectory reweighting

Input : Initial EB parameters $\beta_0 \in [\beta_{\min}, \beta_{\max}]$, $\xi_0 \in \Xi$; learning rates $\eta_\beta, \eta_\xi > 0$; EMA coefficients $\gamma_\beta, \gamma_\xi \in (0, 1]$; constraints ($|\rho| \le \rho_{\max} < 1$, $\eta \ge 0$, fixed k for the banded family).

Output : Streaming EB estimates $\{(\beta_t, \xi_t)\}_{t\ge 0}$ and trajectory weights $w_i^{(t)} \propto \omega_i(\beta_t, \xi_t)$.

for $t = 1, 2, \ldots$ **do**

  Collect batch $\mathcal{B}_t = \{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\}_{i=1}^{G_t}$.

  `// E-step statistics (given` (β_{t-1}, ξ_{t-1})`)`

  foreach $i \in \mathcal{B}_t$ **do**

    $\omega_i^{(t)} \leftarrow \left(L_i^{\beta_{t-1}} s(L_i; \xi_{t-1})\right)^{-1}$.

  $Z_t \leftarrow \sum_{j\in\mathcal{B}_t} \omega_j^{(t)}$; **foreach** $i \in \mathcal{B}_t$ **do**

    $w_i^{(t)} \leftarrow \omega_i^{(t)}/Z_t$.

  $m_t \leftarrow \sum_{i\in\mathcal{B}_t} w_i^{(t)} g_i$; **foreach** $i \in \mathcal{B}_t$ **do**

    $e_i^{(t)} \leftarrow g_i - m_t$.

  $\Lambda_\omega^{(t)} \leftarrow \sum_{i\in\mathcal{B}_t} \omega_i^{(t)}$; $\mathrm{RSS}_\omega^{(t)} \leftarrow \sum_{i\in\mathcal{B}_t} \omega_i^{(t)} (e_i^{(t)})^2$.

  `// M-step: update` β `using Cor. 3.25`

  $g_\beta \leftarrow$

  $$\frac{1}{2}\left[-\sum_{i\in\mathcal{B}_t} \log L_i + \frac{\sum_{i\in\mathcal{B}_t} \omega_i^{(t)} \log L_i}{\Lambda_\omega^{(t)}}\right] + \frac{G_t - 1}{2} \frac{1}{\mathrm{RSS}_\omega^{(t)}} \sum_{i\in\mathcal{B}_t} \omega_i^{(t)} (e_i^{(t)})^2 \log L_i + \partial_\beta \log \pi_\beta(\beta_{t-1}).$$

  $\tilde{\beta}_t \leftarrow \beta_{t-1} + \eta_\beta\, g_\beta$;

  $\beta_t \leftarrow (1 - \gamma_\beta)\beta_{t-1} + \gamma_\beta \,\mathrm{clip}(\tilde{\beta}_t; [\beta_{\min}, \beta_{\max}])$.

  `// M-step: update` ξ `either by ACF-moment (Alg. 2) or EB gradient (Prop. 3.24, Lem. 3.27)`

  if *ACF-moment update* **then**

    $\tilde{\xi}_t \leftarrow \mathrm{ACF_Moment_Fit}\left(\{Y_{i,\tau}\}_{i\in\mathcal{B}_t,\tau}, k\right)$.

  else

    `// EB gradient update`

    $g_\xi \leftarrow \nabla_\xi \ell(\beta_{t-1}, \xi_{t-1})$ evaluated on $\mathcal{B}_t$ via equation 50.

    $\tilde{\xi}_t \leftarrow \xi_{t-1} + \eta_\xi\, g_\xi$.

  $\xi_t \leftarrow (1 - \gamma_\xi)\xi_{t-1} + \gamma_\xi \,\mathrm{proj}_\Xi(\tilde{\xi}_t)$ `// project to` $|\rho| \le \rho_{\max} < 1,\ \eta \ge 0$

  `// Reweight and pass to CAPO update`

  foreach $i \in \mathcal{B}_t$ **do**

    $\omega_i^{(t)} \leftarrow \left(L_i^{\beta_t} s(L_i; \xi_t)\right)^{-1}$.

  $Z_t \leftarrow \sum_{j\in\mathcal{B}_t} \omega_j^{(t)}$; **foreach** $i \in \mathcal{B}_t$ **do**

    $w_i^{(t)} \leftarrow \omega_i^{(t)}/Z_t$.

  Use $\{w_i^{(t)}\}_{i\in\mathcal{B}_t}$ as trajectory weights in the CAPO policy/teacher loss.

  `// Diagnostics and update schedule`

  Clip extreme $w_i^{(t)}$ (e.g., the largest 0.5%); monitor weight entropy and coefficient of variation; optionally update (β_t, ξ_t) only every 2–4 outer steps.

Algorithm 2: ACF-moment estimator for dependence shape (used inside Algorithm 1)

Input : Token-level increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ for each trajectory i; maximum lag k.
Output : Shape estimate $\tilde{\xi} = (\tilde{\rho}, \tilde{\eta}, \dots)$ for the family $s(L; \rho, k, \eta)$ in Definition 3.26.

foreach *trajectory* i **do**
  $\bar{Y}_i \leftarrow \frac{1}{L_i} \sum_{\tau=1}^{L_i} Y_{i,\tau}$;
  for $h = 0, 1, \dots, k$ **do**
    $\hat{\gamma}_i(h) \leftarrow \frac{1}{L_i - h} \sum_{\tau=1}^{L_i - h} (Y_{i,\tau} - \bar{Y}_i)(Y_{i,\tau+h} - \bar{Y}_i)$.

$\bar{\gamma}(h) \leftarrow \frac{\sum_i (L_i - h)\hat{\gamma}_i(h)}{\sum_i (L_i - h)}$ for $h = 0, \dots, k$; $\bar{r}(h) \leftarrow \bar{\gamma}(h)/\bar{\gamma}(0)$ for $h \geq 1$.

$\tilde{\xi} \leftarrow \arg\min_{\rho,\eta} \sum_{h=1}^{k} \left(\bar{r}(h) - \rho^{h^\eta}\right)^2$

`// Optionally downweight large` h `to reduce noise`

Algorithm 3: EB-lite length-only estimator

Algorithm 3 is a lightweight alternative that ignores dependence ($s(L; \xi) \equiv 1$) and fits only the length exponent β. It is based on the approximation

$$\log\left((g_i - m)^2\right) \approx c - \beta \log L_i,$$

which follows from $v_i(\vartheta) \approx \sigma^2 L_i^\beta$ when g_i is centered at $m \approx \mu$. The algorithm:

1. initializes $m^{(0)}$ as the unweighted mean or median of $\{g_i\}$;
2. iteratively regresses $z_i = \log(e_i^2 + \varepsilon)$ on $\log L_i$ to obtain $\hat{\beta}$ from the (negative) slope;
3. reweights trajectories as $w_i \propto L_i^{-\hat{\beta}}$ and updates m as the precision-weighted mean;
4. repeats until convergence of $\hat{\beta}$ and m.

This yields a single-parameter EB estimate $\hat{\beta}$ that can be used as a prior or initialization for the full EB updates with dependence.

Algorithm 4: k-banded covariance weights

Algorithm 4 specializes the general EB weighting rule to the k-banded stretched-geometric dependence model of Definition 3.26. Given a batch $\mathcal{B}_t$, a current length exponent β_t, and dependence parameters (ρ_t, η_t), it:

1. optionally refreshes (ρ_t, η_t) using the ACF-moment estimator (Algorithm 2);
2. computes, for each trajectory i, the shape factor

$$s_i = s(L_i; \rho_t, k, \eta_t) = 1 + \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\rho_t^{h^{\eta_t}}, \qquad m_i = \min\{k, L_i - 1\};$$

3. forms the precision $\omega_i^{(t)} = \left(L_i^{\beta_t} s_i\right)^{-1}$ and normalizes to obtain $w_i^{(t)}$;

Algorithm 3: EB-lite (no dependence): single-parameter length fit

Input : Trajectory pairs $\{(g_i, L_i)\}_{i=1}^{G}$; small regularization constant $\varepsilon > 0$.

Output : Estimate $\hat{\beta}$ and weights $w_i \propto L_i^{-\hat{\beta}}$.

Initialize $m^{(0)} \leftarrow \frac{1}{G}\sum_{i=1}^{G} g_i$ or $m^{(0)} \leftarrow \text{median}\{g_i\}$.

repeat

  for $i = 1, \ldots, G$ **do**

    $e_i \leftarrow g_i - m^{(k)}$; $\quad z_i \leftarrow \log(e_i^2 + \varepsilon)$.

  Fit the regression model $z_i = c - \beta \log L_i + \varepsilon_i$ by (optionally weighted) least squares; set $\hat{\beta}^{(k+1)} \leftarrow -$ (estimated slope).

  Set provisional weights $w_i \propto L_i^{-\hat{\beta}^{(k+1)}}$, then normalize $w_i \leftarrow w_i / \sum_j w_j$.

  Update the aggregator $m^{(k+1)} \leftarrow \sum_{i=1}^{G} w_i g_i$.

until *converged*

Set $\hat{\beta} \leftarrow \hat{\beta}^{(k+1)}$.

4. returns the dependence-corrected weights $\{w_i^{(t)}\}$ and the shape factors $\{s_i\}$.

This gives an explicit implementation of the weighting rule in Proposition 3.14.

Algorithm 5: Joint EB update for β, ρ, and η

Algorithm 5 combines the EB updates for the length exponent β and the k-banded dependence parameters (ρ, η) into a single joint gradient-ascent step on the EB objective $\ell(\beta, \rho, \eta)$, under the stretched-geometric dependence model $s(L; \rho, k, \eta)$. For a given batch $\mathcal{B}_t$, it:

1. computes the E-step statistics $\omega_i, w_i, m_t, e_i, \Lambda_\omega, \text{RSS}_\omega$ under current hyperparameters $(\beta_{t-1}, \rho_{t-1}, \eta_{t-1})$;
2. uses Corollary 3.25 to obtain $g_\beta = \partial_\beta \ell$ and Lemma 3.27 together with Proposition 3.24 to obtain $g_\rho = \partial_\rho \ell$ and $g_\eta = \partial_\eta \ell$;
3. performs projected gradient-ascent steps on β, ρ, η, followed by EMA smoothing with γ_β, γ_ξ;
4. recomputes dependence-corrected weights $w_i^{(t)} \propto [L_i^{\beta_t} s(L_i; \rho_t, k, \eta_t)]^{-1}$ for use in the CAPO update.

This algorithm is the most direct implementation of the EB theory in §3.8 when the practitioner is comfortable optimizing the full parameter vector (β, ρ, η).

Complexity and implementation notes. Per outer step t, computing the EB statistics $\{\omega_i^{(t)}, w_i^{(t)}, m_t, e_i^{(t)}, \Lambda_\omega^{(t)}, \text{RSS}_\omega^{(t)}\}$ has cost $O(G_t)$ for a batch of G_t trajectories. The ACF-moment estimator with maximum lag k has cost $O\Big(\sum_{i \in \mathcal{B}_t} L_i\, k\Big)$, which is typically dominated by the rollout generation cost. In practice, small EMA coefficients $\gamma_\beta, \gamma_\xi \in [0.05, 0.3]$, a bound $|\rho| \leq 0.9$, and clipping of the largest $w_i^{(t)}$ (e.g., at the 99.5th percentile) are sufficient to

Algorithm 4: k-banded covariance weights (stretched-geometric dependence)

Input : Batch $\mathcal{B}_t = \{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\}_{i=1}^{G_t}$;
current length exponent β_t;
current k-banded dependence parameters (ρ_t, η_t);
maximum lag k for the k-band family in Definition 3.26.

Output : Shape factors $s_i = s(L_i; \rho_t, k, \eta_t)$ and trajectory weights
$w_i^{(t)} \propto [L_i^{\beta_t}\, s(L_i; \rho_t, k, \eta_t)]^{-1}$.

`// Optional: refresh` (ρ_t, η_t) `from token-level ACFs`

if *update dependence parameters* **then**

  $\tilde{\xi}_t = (\tilde{\rho}_t, \tilde{\eta}_t) \leftarrow \mathrm{ACF_Moment_Fit}\Big(\{Y_{i,\tau}\}_{i\in\mathcal{B}_t,\tau}, k\Big)$ `// Alg. 2`

  $(\rho_t, \eta_t) \leftarrow \mathrm{proj}_{\Xi}(\tilde{\rho}_t, \tilde{\eta}_t)$

`// Compute` k`-banded shape factors` s_i `and precisions` ω_i

foreach $i \in \mathcal{B}_t$ **do**

  $m_i \leftarrow \min\{k,\, L_i - 1\}$

  $s_i \leftarrow 1 + \dfrac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho_t^{h^{\eta_t}}$ `// Definition 3.26`

  $\omega_i^{(t)} \leftarrow \Big(L_i^{\beta_t}\, s_i\Big)^{-1}$

`// Normalize to obtain Bayes/precision weights`

$Z_t \leftarrow \sum_{j\in\mathcal{B}_t} \omega_j^{(t)}$

foreach $i \in \mathcal{B}_t$ **do**

  $w_i^{(t)} \leftarrow \omega_i^{(t)} / Z_t$

`// Return` k`-banded dependence-corrected weights`

return $\{s_i\}_{i\in\mathcal{B}_t},\ \{w_i^{(t)}\}_{i\in\mathcal{B}_t}$

Algorithm 5: Joint EB update for length β and k-banded dependence (ρ, η)

Input : Batch $\mathcal{B}_t = \{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\}_{i=1}^{G_t}$;
current parameters $\beta_{t-1}, \rho_{t-1}, \eta_{t-1}$;
learning rates $\eta_\beta, \eta_\rho, \eta_\eta > 0$;
EMA $\gamma_\beta, \gamma_\xi \in (0, 1]$;
constraints $\beta \in [\beta_{\min}, \beta_{\max}],\ |\rho| \le \rho_{\max} < 1,\ \eta \ge 0$;
max lag k for $s(L; \rho, k, \eta)$ (Def. 3.26);
priors $\pi_\beta, \pi_\rho, \pi_\eta$.
Output : Updated β_t, ρ_t, η_t and weights $w_i^{(t)} \propto [L_i^{\beta_t} s(L_i; \rho_t, k, \eta_t)]^{-1}$.

`// Optional warm-start for` (ρ, η) `from ACF moments (Alg. 2)`

if *warm-start* **then**
  $(\rho_{t-1}, \eta_{t-1}) \leftarrow \mathrm{proj}_{\Xi}\Big(\mathrm{ACF_Moment_Fit}(\{Y_{i,\tau}\}, k)\Big)$

`// E-step: EB statistics under current` $(\beta_{t-1}, \rho_{t-1}, \eta_{t-1})$

foreach $i \in \mathcal{B}_t$ **do**
  $m_i \leftarrow \min\{k,\ L_i - 1\}$;
  $s_i \leftarrow 1 + \dfrac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho_{t-1}^{h^{\eta_{t-1}}}$;
  $\omega_i \leftarrow \Big(L_i^{\beta_{t-1}}\, s_i\Big)^{-1}$

$\Lambda_\omega \leftarrow \sum_{i \in \mathcal{B}_t} \omega_i$;

foreach $i \in \mathcal{B}_t$ **do**
  $w_i \leftarrow \omega_i / \Lambda_\omega$

$m_t \leftarrow \sum_{i \in \mathcal{B}_t} w_i g_i$;

foreach $i \in \mathcal{B}_t$ **do**
  $e_i \leftarrow g_i - m_t$

$\mathrm{RSS}_\omega \leftarrow \sum_{i \in \mathcal{B}_t} \omega_i e_i^2$

`// M-step: gradients of EB objective` $\ell(\beta, \rho, \eta)$

Compute $g_\beta = \partial_\beta \ell$ from Cor. 3.25 (using current $\{\omega_i, e_i, L_i\}$)

Compute $g_\rho = \partial_\rho \ell$ and $g_\eta = \partial_\eta \ell$ from Prop. 3.24 and Lem. 3.27 (using current $\{\omega_i, e_i, s_i\}$)

`// Gradient-ascent updates with projection and EMA`

$\tilde{\beta}_t \leftarrow \beta_{t-1} + \eta_\beta\, g_\beta$;$\tilde{\rho}_t \leftarrow \rho_{t-1} + \eta_\rho\, g_\rho$;$\tilde{\eta}_t \leftarrow \eta_{t-1} + \eta_\eta\, g_\eta$;

$\beta_t \leftarrow (1 - \gamma_\beta)\beta_{t-1} + \gamma_\beta\, \mathrm{clip}(\tilde{\beta}_t; [\beta_{\min}, \beta_{\max}])$;

$\rho_t \leftarrow (1 - \gamma_\xi)\rho_{t-1} + \gamma_\xi\, \mathrm{clip}(\tilde{\rho}_t; [-\rho_{\max}, \rho_{\max}])$;

$\eta_t \leftarrow (1 - \gamma_\xi)\eta_{t-1} + \gamma_\xi\, \max\{\tilde{\eta}_t,\ 0\}$;

`// Recompute` k`-banded weights at` $(\beta_t, \rho_t, \eta_t)$ `for CAPO`

foreach $i \in \mathcal{B}_t$ **do**
  $m_i \leftarrow \min\{k,\ L_i - 1\}$;
  $s_i \leftarrow 1 + \dfrac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho_t^{h^{\eta_t}}$;
  $\omega_i \leftarrow \Big(L_i^{\beta_t}\, s_i\Big)^{-1}$

$\Lambda_\omega \leftarrow \sum_{j \in \mathcal{B}_t} \omega_j$;

foreach $i \in \mathcal{B}_t$ **do**
  $w_i^{(t)} \leftarrow \omega_i / \Lambda_\omega$

return $\beta_t, \rho_t, \eta_t, \{w_i^{(t)}\}_{i \in \mathcal{B}_t}$

prevent any single trajectory from dominating the update.

Numerical stability, identifiability, and practical defaults. In RLVR regimes with long-context CoT, the raw length factors $L_i^{-\beta}$ can underflow for large L_i and moderate β. A numerically stable implementation should therefore compute weights in log-space: $\log \omega_i = -\beta \log L_i - \log s(L_i; \xi)$ and then use a log-sum-exp reduction to obtain $\Lambda_\omega = \sum_i \omega_i$. Similarly, when the dependence model includes exponential terms (e.g., $e^{-L/\xi}$), it is preferable to evaluate $\log s(L; \xi)$ directly.

From an identifiability perspective, (β, ξ) are only weakly identifiable in small batches when the empirical length distribution is narrow. In such cases, we recommend (i) restricting β to a conservative interval (e.g., $[0, 1]$), (ii) using the outcomes-only EB objective as a warm-start for the full CAPO-EB updates, and (iii) employing mild regularization or informative priors as in §3.4. These steps stabilize estimation without altering the unbiasedness constraint in equation 6.

5 Experiments

We evaluate CAPO on a controlled RLVR setting designed to isolate the effect of advantage aggregation. All results are produced with a single open-weight model (Qwen2.5–1.5B–Instruct) and a single task distribution (CountDown).

5.1 Setup

Task. CountDown presents a target integer and a small multiset of integers; the model must return an arithmetic expression that evaluates to the target. We use the standard exact-match verifier, yielding a binary outcome reward.

Model. We finetune Qwen2.5–1.5B–Instruct with group-based on-policy RL. For each prompt, we sample $N = 8$ trajectories per update, matching the aggregation dimension assumed by GRPO-style estimators.

Context budgets. We report two response-length regimes: (i) a *short* regime with maximum response length 2048 tokens and (ii) a *long* regime with maximum response length 8192 tokens. Unless otherwise noted, all other hyperparameters are kept identical across regimes.

Implementation. We implement all methods within the same training stack (VERL), holding fixed the rollout procedure, optimizer, and training schedule. This ensures that differences can be attributed to the advantage aggregation / normalization rule rather than to unrelated systems choices.

Group-based objective and estimator. All baselines and CAPO variants use the same group-based on-policy objective. Concretely, for each prompt we sample N trajectories, compute token-level policy-gradient terms using the same clipping and advantage conventions, and

Hyperparameter	Value
Base policy model	Qwen2.5–1.5B–Instruct
RL stack	VERL
Task / reward	CountDown verifier (binary success)
Trajectories per prompt (N)	8
Max response tokens	$\{512, 1024, 2048\}$
Optimizer	—
Learning rate	—
PPO/GRPO cliprange	—
Rollout temperature	—
Prompts per update (M)	—
Evaluation frequency	—
Random seeds	—

Table 2 Key optimization and rollout hyperparameters used in the CountDown RLVR experiments. We hold these fixed across methods within a given context budget; only the advantage aggregation / normalization rule varies.

differ only in how the per-trajectory advantages (and hence token-level losses) are reweighted as a function of trajectory length and dependence. This isolates the effect of the aggregation rule discussed in §2–§3.

Hyperparameters and reproducibility. We use identical optimizer, rollout, and training-schedule hyperparameters across all methods within a given context budget. For transparency and reproducibility, we report the key optimization hyperparameters in Table 2. Unless otherwise stated, all reported curves are averaged over multiple random seeds; the stability metrics in Table 4 are computed from the per-seed validation trajectories prior to averaging.

5.2 Baselines and method variants

We compare CAPO against a set of aggregation baselines that progressively add or remove normalizations:

- **DAPO normalization** (token-mean): GRPO with token-averaged policy loss.
- **GRPO normalization** (seq-mean-token-mean): GRPO with per-sequence normalization of the loss, which empirically reduces length bias but can increase variance.
- **Dr. GRPO normalization:** GRPO with distributional-reweighting correction (Dr. GRPO) combined with the GRPO normalization.
- ΔL **normalization** ($\alpha \in \{0.75, 1.0\}$): length-weighted normalization via GRPO++-style trajectory weighting.

We report CAPO with outcome-only trajectory covariance estimation: **CAPO-EB-lite** (diagonal empirical Bayes) and **CAPO-EB** (banded empirical Bayes).

Method	CountDown (1.5B, 2048)	CountDown (1.5B, 8192)
GRPO Norm	—	—
DAPO Norm	—	—
Dr. GRPO Norm	—	—
ΔL ($\alpha = 1$)	—	—
ΔL ($\alpha = 0.75$)	—	—
CAPO–EB (best)	—	—

Table 3 Main accuracy on CountDown with Qwen2.5–1.5B–Instruct. We report Avg@8 under a short (2048) and long (8192) response budget.

Method	Monotonicity ↑	NDR ↓	TTT ↓
ΔL (best)	—	—	—
CAPO–EB (best)	—	—	—

Table 4 Stability and efficiency diagnostics computed from validation curves. Higher monotonicity is better; lower NDR and lower TTT are better.

5.3 Metrics

Accuracy. For each validation prompt, we sample $N = 8$ trajectories and report (i) *Avg@8*, the average verifier success over trajectories, and (ii) *Best@8*, the success rate of the best trajectory within each group (equivalently, pass@8 under a binary verifier).

Stability and efficiency. We summarize learning-curve stability and sample efficiency with:

- **Monotonicity:** the Pearson correlation between evaluation step and validation Avg@8.
- **Negative drift rate (NDR):** the fraction of evaluation intervals with a decrease in validation Avg@8.
- **Tokens-to-target (TTT):** the cumulative number of training tokens processed until validation Avg@8 first exceeds a fixed target threshold.

5.4 Main results

Table 3 reports end performance under both context budgets. Figure 3 shows representative learning dynamics.

5.5 Stability and efficiency

Table 4 aggregates stability and efficiency diagnostics. Figure 4 visualizes the stability trade-off across methods.

5.6 Length equity

To make length effects explicit, we stratify validation by response-length deciles and report verifier success within each stratum.

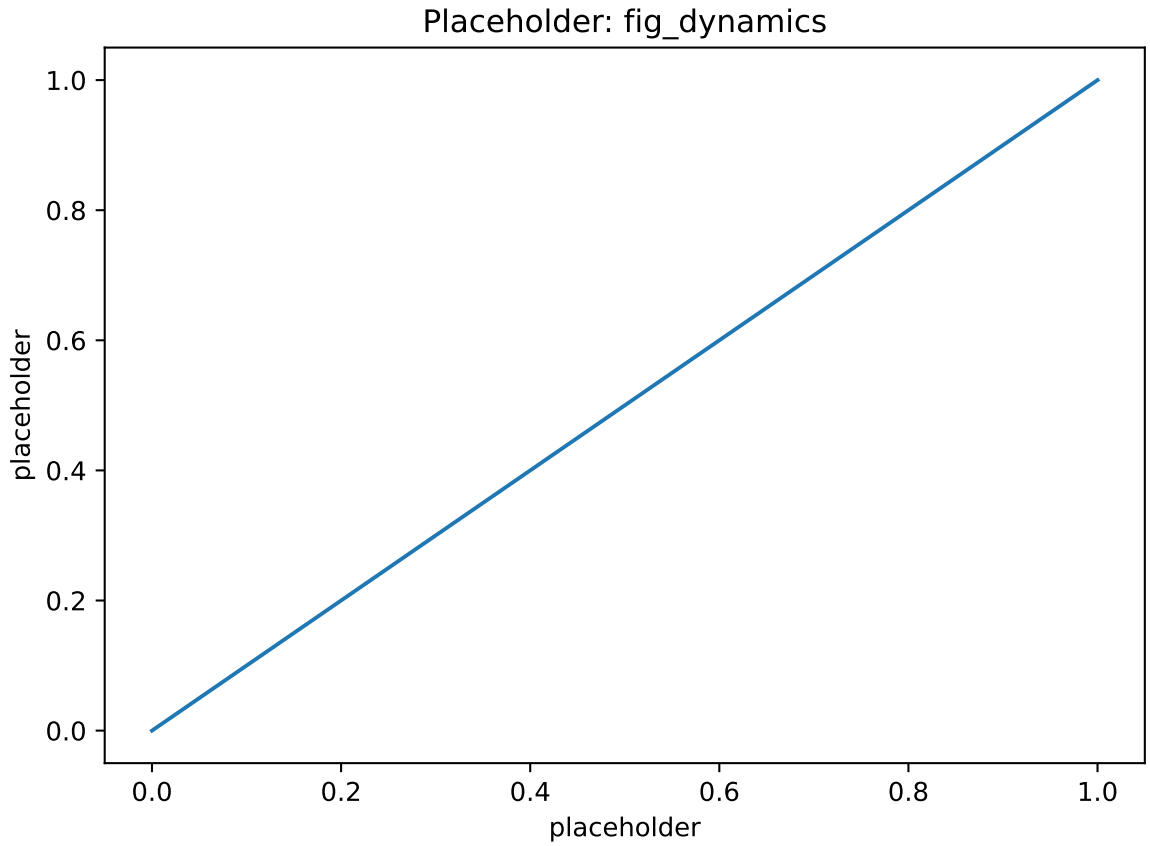

Figure 3 Training dynamics (validation Avg@8 vs. global step) for a representative subset of methods.

5.7 Additional diagnostics and failure modes

Weight degeneracy. A central practical concern in any variance-based weighting scheme is *degeneracy* of the normalized weights w_i: if one trajectory receives nearly all mass, the effective number of trajectories contributing to the update collapses. To monitor this, we recommend reporting the effective sample size $\mathrm{ESS} := (\sum_i w_i)^2/(\sum_i w_i^2)$ and/or the entropy $-\sum_i w_i \log w_i$ of the normalized weights. CAPO includes explicit clipping and ε-regularization steps (Algorithms 3–1) to prevent pathological collapse.

Length-conditioned bias. Length-based normalization can introduce systematic differences in how short and long trajectories are reinforced. The length-decile analysis in Figure 5 is intended to expose such effects. For publication-quality reporting, we recommend additionally stratifying by reward outcome (success/failure) and by prompt difficulty to distinguish genuine credit assignment from spurious length preferences.

Model misspecification. The variance law $v(L) \propto L^{\beta} s(L;\xi)$ is a working model. When this model is misspecified, CAPO remains unbiased (because equation 6 enforces unbiasedness), but suboptimal weighting can increase variance and exacerbate instability. The robustness

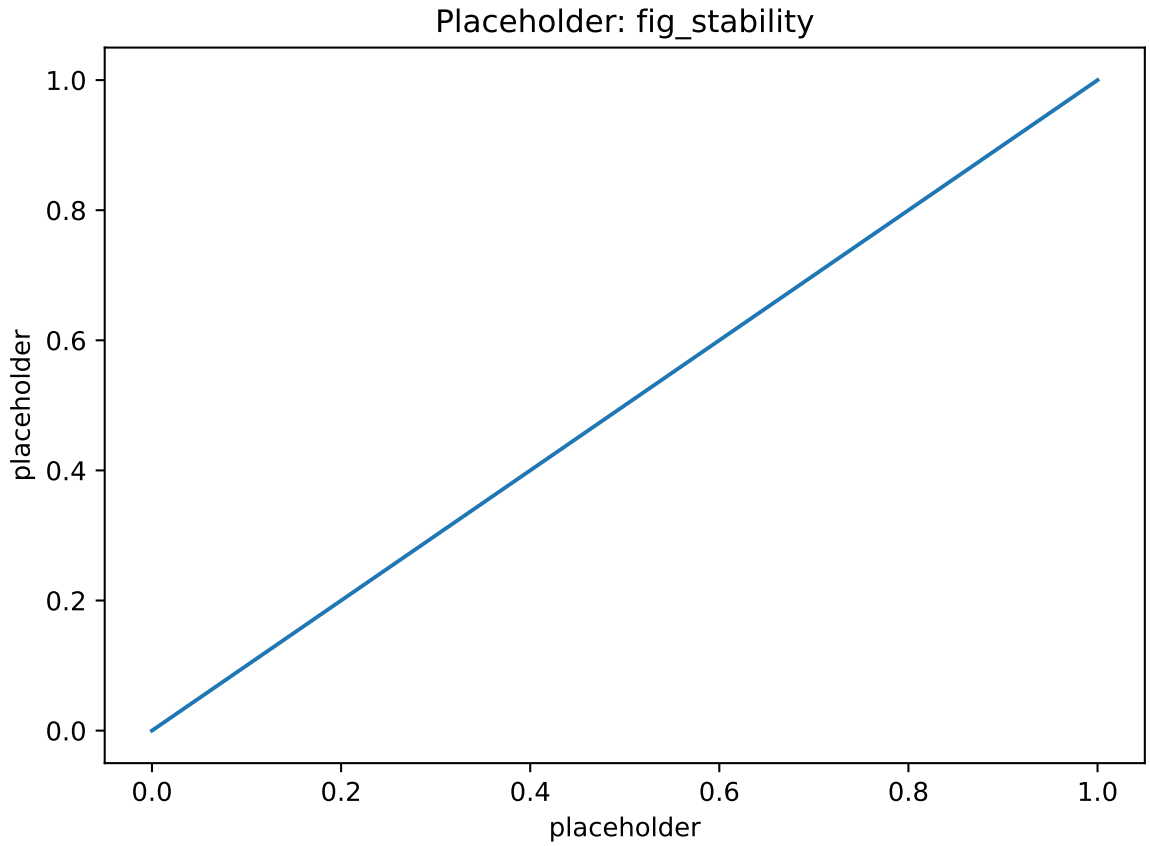

Figure 4 Stability summary. Each point corresponds to a method, plotted by monotonicity (higher is better) and negative drift rate (lower is better).

results in §3.9 and the covariance examples in Appendix D provide guidance for diagnosing when simple dependence surrogates (e.g., exponential or compound-symmetry) are adequate.

6 Conclusion

This paper developed *Covariance-Aware Policy Optimization (CAPO)*, a variance-aware aggregation principle for policy-gradient updates under the length heterogeneity and within-trajectory dependence that arise in reinforcement learning with verifiable rewards (RLVR). Starting from a minimum-variance unbiased estimation (MVUE) formulation of trajectory reweighting, we showed that inverse-variance weights arise as the unique optimal linear unbiased rule (Lemma 3.6) and that classical length-normalizations such as ΔL correspond to special cases when trajectory variance grows proportionally to length. We then introduced a flexible heteroscedastic Gaussian working model with structured dependence through a *shape factor* $s(L;\xi)$, established a posterior-precision view of optimal weighting with an invariant Jeffreys baseline, and derived principled sublinear length exponents $\beta < 1$ under correctness-linked variance reduction. To make these ideas operational, we proposed empirical-Bayes (EB) estimators for (β, ξ), provided asymptotic guarantees under weak dependence, and gave

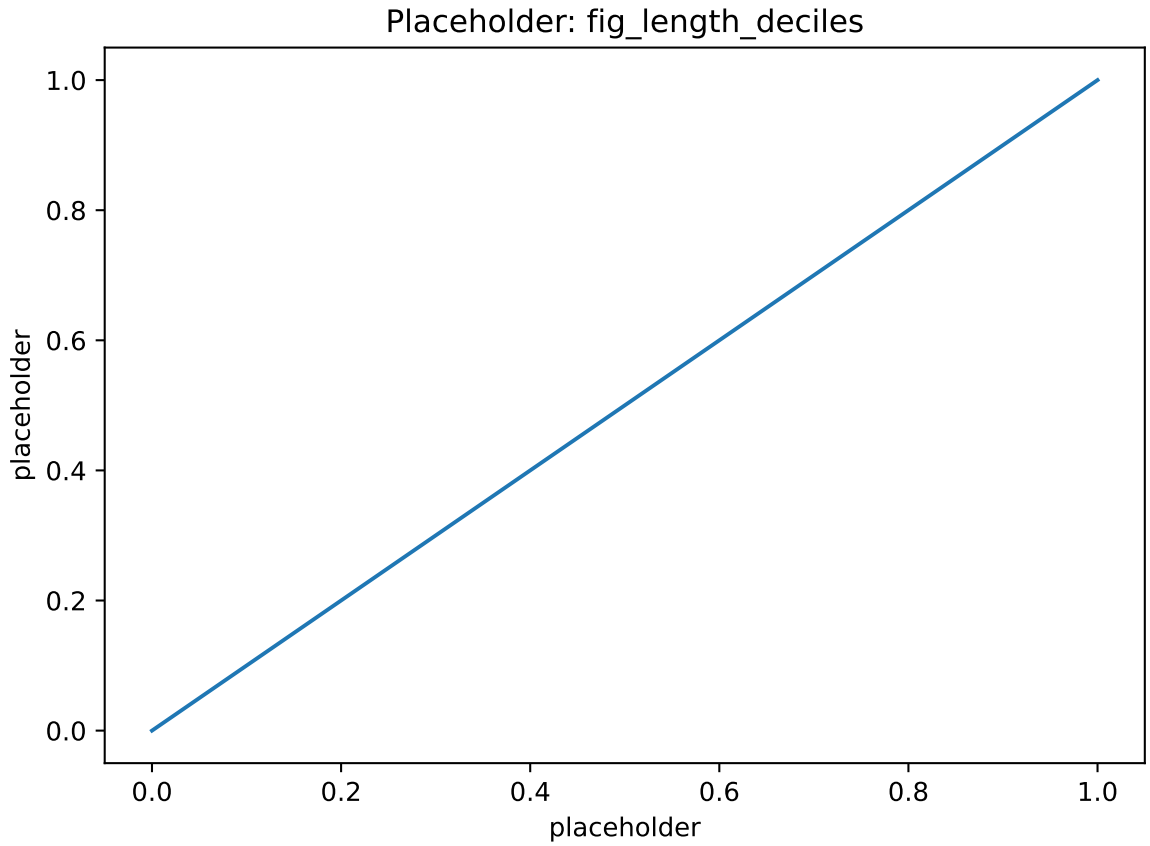

Figure 5 Verifier success by response-length decile. This analysis highlights whether a method disproportionately favors short (or long) trajectories even when end-to-end accuracy is similar.

implementable CAPO variants that interpolate between outcomes-only weighting and full token-level dependence corrections.

Beyond the specific RLVR setting, the CAPO perspective clarifies a general design principle for stochastic-gradient aggregation with variable-length or variable-cost samples: *the correct normalization is determined by how the sample-level covariance grows with length and dependence, rather than by length alone.* In this sense, CAPO connects the practical engineering of stable LLM RL training to classical ideas from weighted least squares, generalized least squares, and empirical Bayes.

Limitations and future directions. CAPO is intentionally conservative in its modeling assumptions: we treat the within-trajectory covariance through low-dimensional summaries (bandwidth, decay, or compound-symmetry surrogates) and adopt cross-trajectory uncorrelatedness as a working approximation (Assumption 2.1). Two directions appear especially important for future work. First, prompt-level random effects and group-based baselines can induce non-negligible dependence across trajectories within a batch; extending CAPO to explicit block-structured cross-trajectory covariance is a natural next step. Second, policy-gradient contributions may exhibit heavy tails or non-Gaussianity under rare but extreme

rewards; robustness to tail behavior and stable estimation of variance under such distributions merit dedicated study. We view these extensions as complementary to the present contribution: CAPO provides a rigorous baseline framework upon which richer covariance models and robust estimators can be layered without sacrificing unbiasedness.

7 Acknowledgment

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A Proof of the Minimum-Variance Unbiased (MVU) Weights

Lemma A.1 (Inverse-variance MVU weights)**.** *Assume* $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) = \mathbf{0}$ *for* $i \neq j$ *and let* $v_i := \Psi\big(\mathrm{Var}(\boldsymbol{g}_i)\big) \in (0, \infty)$ *for a fixed positive, linear, monotone functional* Ψ *on* $\mathbb{S}^p_+$ *(e.g.,* $\Psi(\boldsymbol{A}) = \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}$ *with* $\|\boldsymbol{u}\|{=}1$*, or* $\Psi(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A})$*). Among all linear unbiased estimators* $\widehat{\boldsymbol{g}} = \sum_{i=1}^G x_i \boldsymbol{g}_i$ *satisfying* $\sum_{i=1}^G x_i = 1/M$*, the weights*

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^G v_j^{-1}} \tag{10}$$

uniquely minimize the dispersion $\Psi\big(\mathrm{Var}(\widehat{\boldsymbol{g}})\big)$*.*

Proof. By linearity of covariance and the cross-covariance assumption,

$$\mathrm{Var}(\widehat{\boldsymbol{g}}) = \mathrm{Var}\Big(\sum_{i=1}^G x_i \boldsymbol{g}_i\Big) = \sum_{i=1}^G x_i^2\,\mathrm{Var}(\boldsymbol{g}_i).$$

Applying Ψ (positive, linear, monotone on $\mathbb{S}^p_+$),

$$\Psi\big(\mathrm{Var}(\widehat{\boldsymbol{g}})\big) = \sum_{i=1}^G x_i^2\,\Psi\big(\mathrm{Var}(\boldsymbol{g}_i)\big) = \sum_{i=1}^G v_i\,x_i^2.$$

Thus the problem reduces to the strictly convex quadratic program

$$\min_{\boldsymbol{x}} \sum_{i=1}^G v_i x_i^2 \quad \text{s.t.} \quad \sum_{i=1}^G x_i = \frac{1}{M},$$

with $v_i > 0$. The Lagrangian is $\mathcal{L}(\boldsymbol{x}, \lambda) = \sum_i v_i x_i^2 - \lambda\Big(\sum_i x_i - \frac{1}{M}\Big)$. First-order optimality yields $2v_i x_i - \lambda = 0$, i.e.

$$x_i = \frac{\lambda}{2v_i}.$$

Imposing the constraint,

$$\sum_{i=1}^G x_i = \frac{\lambda}{2}\sum_{i=1}^G \frac{1}{v_i} = \frac{1}{M} \quad \Longrightarrow \quad \lambda = \frac{2}{M\sum_{j=1}^G v_j^{-1}}.$$

Substituting back gives

$$x_i^\star = \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^G v_j^{-1}},$$

which is the unique minimizer by strict convexity. □

Remark A.2. Taking $\Psi(\boldsymbol{A}) = \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}$ recovers the familiar "inverse-variance" rule with $v_i = \boldsymbol{u}^\top \mathrm{Var}(\boldsymbol{g}_i)\boldsymbol{u}$. Taking $\Psi = \mathrm{tr}$ yields the same weights with $v_i = \mathrm{tr}\,\mathrm{Var}(\boldsymbol{g}_i)$. If $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) \neq \mathbf{0}$ for some $i \neq j$, the optimal weights depend on the full covariance matrix and are not generally diagonal inverse-variance.

B ΔL under Token Independence

This appendix provides an explicit, self-contained derivation of the ΔL warm-up case (token independence) and a notation bridge to He et al. (2025). The main text in §3 states the result succinctly; here we record the derivation and the exact conditions under which the inverse-length rule is MVU.

Under an i.i.d. token model with $\mathrm{Var}(\boldsymbol{Z}_{i,t}) = \boldsymbol{\Sigma}_0$ and $\boldsymbol{\Gamma}_i(k) = \mathbf{0}$ for $k \neq 0$,

$$\mathrm{Var}(\boldsymbol{g}_i) = L_i\, \boldsymbol{\Sigma}_0 \quad \Longrightarrow \quad v_i \propto L_i. \tag{58}$$

Substituting into the MVU weights equation 10 yields $x_i \propto L_i^{-1}$, i.e., the ΔL rule with exponent $\alpha = 1$. More generally, the practical one-parameter family

$$x_i \;\propto\; L_i^{-\alpha}, \qquad \alpha \in [0, 1], \tag{59}$$

interpolates between constant weighting ($\alpha = 0$) and inverse-length weighting ($\alpha = 1$), providing a baseline that CAPO formalizes via posterior precisions in §3.

C Bartlett summation identity for trajectory covariance

Lemma C.1 (Bartlett summation identity for trajectory covariance)**.** *Assume second–order stationarity along the token index for trajectory i: for all t and all lags $k \in \mathbb{Z}$,*

$$\mathbb{E}\, \boldsymbol{Z}_{i,t} = \boldsymbol{m}_i \quad \textit{(constant)}, \qquad \mathrm{Cov}(\boldsymbol{Z}_{i,t}, \boldsymbol{Z}_{i,t-k}) = \boldsymbol{\Gamma}_i(k).$$

Let $\boldsymbol{g}_i = \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}$. Then

$$\mathrm{Var}(\boldsymbol{g}_i) \;=\; \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k). \tag{8}$$

Proof. Write $\widetilde{\boldsymbol{Z}}_{i,t} := \boldsymbol{Z}_{i,t} - \boldsymbol{m}_i$. Then

$$\mathrm{Var}(\boldsymbol{g}_i) = \mathrm{Cov}\Big(\sum_{t=1}^{L_i} \widetilde{\boldsymbol{Z}}_{i,t}, \sum_{s=1}^{L_i} \widetilde{\boldsymbol{Z}}_{i,s}\Big) = \sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \mathrm{Cov}\big(\widetilde{\boldsymbol{Z}}_{i,t}, \widetilde{\boldsymbol{Z}}_{i,s}\big).$$

By stationarity, $\mathrm{Cov}(\widetilde{\boldsymbol{Z}}_{i,t}, \widetilde{\boldsymbol{Z}}_{i,s}) = \boldsymbol{\Gamma}_i(t-s)$. Hence

$$\mathrm{Var}(\boldsymbol{g}_i) = \sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \boldsymbol{\Gamma}_i(t-s).$$

Group terms by the lag $k = t - s$. For a fixed k with $|k| \leq L_i - 1$, the number of index pairs (t, s) in $\{1, \ldots, L_i\}^2$ satisfying $t - s = k$ is $L_i - |k|$ (for $k \geq 0$: $s = 1, \ldots, L_i - k$; for $k < 0$: $t = 1, \ldots, L_i - |k|$). Therefore

$$\sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \boldsymbol{\Gamma}_i(t-s) = \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k),$$

which is equation 8. □

D Illustrative examples

In this section we give two concrete examples illustrating the EB–CAPO algorithm: (i) a simple hand-computable toy example that makes the weighting mechanism transparent, and (ii) a simulated rollout experiment that can be used to compare CAPO with a GRPO-style baseline and empirically demonstrate the variance and sample-efficiency gains predicted by the theory.

D.0.1 Toy example: three trajectories with different lengths

Consider a single batch of $G = 3$ trajectories with lengths

$$L_1 = 16, \qquad L_2 = 64, \qquad L_3 = 256.$$

Suppose their scalarized returns (e.g., episode rewards) are

$$g_1 = 0.8, \qquad g_2 = 1.0, \qquad g_3 = 1.1.$$

We will assume, for this toy example, that the dependence factor is constant, $s(L; \xi) \equiv 1$, so that

$$\omega_i(\beta, \xi) = \omega_i(\beta) = L_i^{-\beta}$$

and we focus on how the length exponent β changes the weights and the aggregated estimate of μ.

Case 1: GRPO-style ΔL weighting ($\beta = 1$). Setting $\beta = 1$ and $s \equiv 1$ corresponds to the ΔL rule discussed in Proposition 3.29. The (unnormalized) precisions are

$$\omega_1 = L_1^{-1} = 16^{-1} = 0.0625, \quad \omega_2 = 64^{-1} \approx 0.015625, \quad \omega_3 = 256^{-1} \approx 0.00390625.$$

The normalization constant is

$$\Lambda_\omega = \omega_1 + \omega_2 + \omega_3 = 0.0625 + 0.015625 + 0.00390625 = 0.08203125.$$

Hence the normalized weights are

$$w_1 = \frac{\omega_1}{\Lambda_\omega} \approx 0.7619, \quad w_2 \approx 0.1905, \quad w_3 \approx 0.0476.$$

The aggregated estimate is

$$m(\beta = 1) = \sum_{i=1}^{3} w_i g_i \approx 0.7619 \cdot 0.8 + 0.1905 \cdot 1.0 + 0.0476 \cdot 1.1 \approx 0.846.$$

Here the shortest trajectory gets the largest weight, because GRPO implicitly assumes that longer episodes are noisier (variance growing proportionally with length).

Case 2: EB-favored $\beta < 1$ when longer rollouts are less noisy. Suppose the empirical EB fit (e.g., via Algorithm 3) finds that the residual variance decreases approximately like $L^{-0.5}$, corresponding to an optimal $\beta^\star \approx 0.5$. Then

$$\omega_1 = L_1^{-1/2} = 16^{-1/2} = 0.25, \quad \omega_2 = 64^{-1/2} = 0.125, \quad \omega_3 = 256^{-1/2} = 0.0625,$$

and

$$\Lambda_\omega = 0.25 + 0.125 + 0.0625 = 0.4375.$$

The normalized weights become

$$w_1 = \frac{0.25}{0.4375} \approx 0.5714, \quad w_2 \approx 0.2857, \quad w_3 \approx 0.1429.$$

The aggregated estimate is now

$$m(\beta = 0.5) \approx 0.5714 \cdot 0.8 + 0.2857 \cdot 1.0 + 0.1429 \cdot 1.1 \approx 0.892.$$

Relative to the GRPO-style $\beta = 1$ case, EB has shifted mass from the shortest trajectory towards the longer, more stable trajectories, because the data indicated that per-token noise decreases with length. The toy example makes the effect of β on the weights and on the overall estimator $m(\beta, \xi)$ transparent, while keeping all computations fully explicit.

Case 3: EB-favored $\beta > 1$ when very long rollouts are degraded. Conversely, if the empirical analysis (e.g., based on RSS_ω across length buckets) suggests that very long rollouts are degraded due to reward sparsity or drift, the EB objective $\ell(\beta, \xi)$ will favor $\beta > 1$, further downweighting extremely long episodes relative to the $\beta = 1$ baseline. This case can be illustrated by repeating the above calculations with, say, $\beta = 1.5$, in which case L_3 receives much smaller weight than in the GRPO configuration.

These hand-computable examples are convenient for visual demonstrations (e.g., plotting the weights $w_i(\beta)$ as a function of β) and for sanity-checking code paths that implement Algorithm 1.

D.0.2 Simulated rollouts: CAPO–EB versus GRPO

We now outline a fully specified simulated experiment that can be used to compare CAPO with a GRPO-style baseline on a controlled synthetic task. The goal is to isolate the effect of length- and covariance-aware weighting on the variance and sample-efficiency of the training signal.

Generative model. We construct a synthetic rollout generator with the following components:

- **Length distribution.** At each outer step t, draw G_t trajectory lengths i.i.d. from a heavy-tailed discrete distribution to mimic realistic RLHF setups, for example

  $$L_i \sim \mathrm{Uniform}\{32, 64, 128, 256\} \quad \text{or} \quad L_i \sim \text{Discrete-Lognormal}(\mu_L, \sigma_L^2) \text{ clipped to } [16, 512].$$

- **Token-level increments with dependence.** For each trajectory i, generate scalar increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ according to a Gaussian AR(1) process with length-dependent variance:

$$Y_{i,1} \sim \mathcal{N}(0, \sigma_0^2),$$
$$Y_{i,\tau} \mid Y_{i,\tau-1} \sim \mathcal{N}\Big(\rho\, Y_{i,\tau-1},\, \sigma_\epsilon^2 L_i^{-\gamma}\Big), \qquad \tau = 2, \ldots, L_i,$$

  where $\rho \in (0,1)$ controls temporal dependence and $\gamma > 0$ controls how per-token noise decays with length. For large L_i and fixed ρ, γ, this construction induces an approximate variance for the trajectory average that behaves like $L_i^{-\gamma}$ up to a multiplicative factor depending on ρ.

- **Scalarized return.** Define the scalarized trajectory return as

$$g_i = \mu^\star + \frac{1}{L_i} \sum_{\tau=1}^{L_i} Y_{i,\tau} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{tail}}^2),$$

  where $\mu^\star$ is the true global mean (e.g., the "teacher" target) and ε_i is an additional independent noise term capturing unmodeled effects. By construction, conditional on L_i and $\{Y_{i,\tau}\}$, we have $\mathbb{E}[g_i \mid L_i] = \mu^\star$ and $\mathrm{Var}(g_i \mid L_i) \approx \sigma^2 L_i^{-\gamma}$ for a suitable σ^2, so the true length–variance exponent is $\beta^\star \approx -\gamma$ (under the family $v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i; \xi)$, we have $\beta^\star = -\gamma$ and $s \equiv 1$ in the simplest case).

Compared estimators (CAPO–EB vs. GRPO). On top of this generator, we compare two weighting strategies for estimating $\mu^\star$ (or, in an RL context, for aggregating policy gradients):

- **GRPO-style baseline.** Use the ΔL rule (Proposition 3.29) regardless of the actual noise law:

$$\omega_i^{\text{GRPO}} = L_i^{-1}, \qquad w_i^{\text{GRPO}} = \frac{\omega_i^{\text{GRPO}}}{\sum_j \omega_j^{\text{GRPO}}}, \qquad m_{\text{GRPO}} = \sum_i w_i^{\text{GRPO}} g_i.$$

  In a full GRPO implementation, these same weights would be used to aggregate per-trajectory gradients or log-prob increments for the policy update.

- **CAPO–EB.** Use the EB objective $\ell(\beta, \xi)$ in equation 48 to fit β (and optionally ξ), and then weight trajectories according to

$$\omega_i^{\text{CAPO}} = L_i^{-\beta_t} s(L_i; \xi_t)^{-1}, \qquad w_i^{\text{CAPO}} = \frac{\omega_i^{\text{CAPO}}}{\sum_j \omega_j^{\text{CAPO}}}, \qquad m_{\text{CAPO}} = \sum_i w_i^{\text{CAPO}} g_i.$$

  Two variants are particularly simple to implement:

  - *CAPO–EB-lite:* Set $s \equiv 1$ and estimate β_t using Algorithm 3 (no token-level ACFs required).
  - *Full CAPO–EB:* Use $s(L; \rho, k, \eta)$ from Definition 3.26, update β_t via Corollary 3.25, and update $\xi_t = (\rho_t, \eta_t)$ via Algorithm 1 (possibly combined with the ACF-moment estimator, Algorithm 2).

Simulation protocol. A simple protocol that is straightforward to implement and plot is:

1. Fix hyperparameters: true $\mu^\star$, ρ, γ, σ_0^2, σ_ϵ^2, σ_{tail}^2, length distribution for L_i, batch size G_t, and number of outer steps T.
2. For each outer step $t = 1, \ldots, T$:
   (a) Generate a batch $\mathcal{B}_t$ using the generative model above.
   (b) **GRPO branch:** Compute $(w_i^{\text{GRPO}}, m_{\text{GRPO}}^{(t)})$ using $\omega_i^{\text{GRPO}} = L_i^{-1}$.
   (c) **CAPO branch:** Run one iteration of Algorithm 1 (or EB-lite, Algorithm 3) to update (β_t, ξ_t) and compute $(w_i^{\text{CAPO}}, m_{\text{CAPO}}^{(t)})$.
3. Repeat the entire T-step process over R independent trials (e.g., $R = 500$ or $R = 1000$), re-simulating fresh data each time, and record the sequences $\{m_{\text{GRPO}}^{(t,r)}\}_{t,r}$ and $\{m_{\text{CAPO}}^{(t,r)}\}_{t,r}$.

Evaluation metrics. On this synthetic task the ground truth $\mu^\star$ is known, which allows clean comparisons:

- **Mean-squared error (MSE) over time.** For each method $M \in \{\text{GRPO}, \text{CAPO}\}$, define

$$\text{MSE}_M(t) = \frac{1}{R} \sum_{r=1}^{R} \left(m_M^{(t,r)} - \mu^\star \right)^2.$$

  Plotting $\text{MSE}_{\text{GRPO}}(t)$ and $\text{MSE}_{\text{CAPO}}(t)$ on the same axis typically shows that CAPO reaches a given error threshold (e.g., 10^{-2}) in many fewer outer steps, reflecting its variance-optimal weighting.

- **Variance and bias decomposition.** Because both estimators are unbiased in the model of Section 3.8, the MSE is dominated by variance, $\text{MSE}_M(t) \approx \text{Var}(m_M^{(t)})$. One can empirically confirm this by computing

$$\widehat{\text{Bias}}_M(t) = \frac{1}{R} \sum_{r=1}^{R} m_M^{(t,r)} - \mu^\star,$$

  and checking that $|\widehat{\text{Bias}}_M(t)|$ is negligible compared to $\sqrt{\text{MSE}_M(t)}$.

- **Effective sample size (ESS).** For each method M at step t, define

$$\text{ESS}_M(t) = \frac{1}{\sum_{i \in \mathcal{B}_t} (w_i^M)^2},$$

  where w_i^M are the normalized weights (either GRPO or CAPO). CAPO typically yields a larger ESS in regimes where GRPO over- or under-weights certain length ranges, indicating more efficient usage of the same batch of trajectories.

Integrating with a policy update. The above simulation evaluates CAPO and GRPO purely as estimators of $\mu^\star$ (i.e., of the scalarized training signal). To embed this into a full policy-learning loop, one can replace the role of g_i by per-trajectory gradient estimates (or log-prob increments) and use the same weights w_i^{GRPO} vs. w_i^{CAPO} to aggregate them. Because the EB estimator $m(\beta, \xi)$ has strictly smaller variance than any fixed-weight scheme of the form $w_i \propto L_i^{-\beta_0}$ when the working model is approximately correct, the same simulation framework can then be used to compare downstream learning curves (e.g., average return or teacher score versus wall-clock steps) and empirically demonstrate the superiority of CAPO over GRPO on this controlled benchmark.